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Prakash et al.

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(54) **MULTI-STAGE PROGRAMMING TECHNIQUES WITH THREE STATES PER MEMORY CELL PARITY**

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(60) Provisional application No. 63/434,293, filed on Dec. 21, 2022.

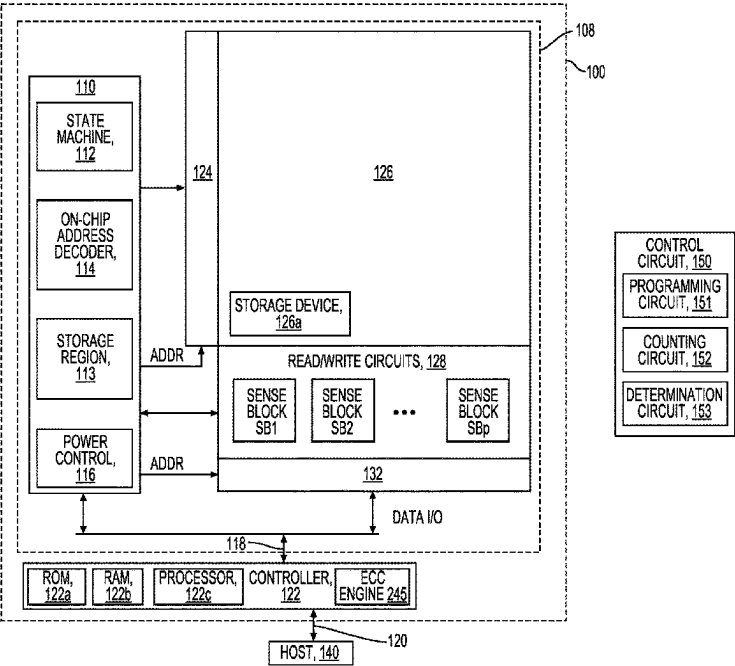
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G11C 16/04 (2006.01)
G11C 11/56 (2006.01)
G11C 16/08 (2006.01)
G11C 16/10 (2006.01)
G11C 16/34 (2006.01)
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CPC **G11C 16/3459** (2013.01); **G11C 16/08** (2013.01); **G11C 16/102** (2013.01)

(58) **Field of Classification Search**
CPC ... G11C 16/3459; G11C 16/08; G11C 16/102; G11C 11/5628; G11C 16/0483; G11C 16/10
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(57) **ABSTRACT**
The memory device includes a memory block with an array of memory cells arranged word lines. The memory device also includes control circuitry that is configured to program final data into a selected word line in a multi-pass programming operation that includes a first pass and a second pass. In the first pass, the control circuitry is configured to program the memory cells of the selected word line to foggy data and program parity data in the memory device. The parity data includes three possible data states. Prior to the second pass, the control circuitry is configured to read the foggy data and the parity data and reconstruct the final data from the foggy data and the parity data. In the second pass, the control circuitry is configured to program the memory cells of the selected word line from the foggy data to the final data.

20 Claims, 18 Drawing Sheets



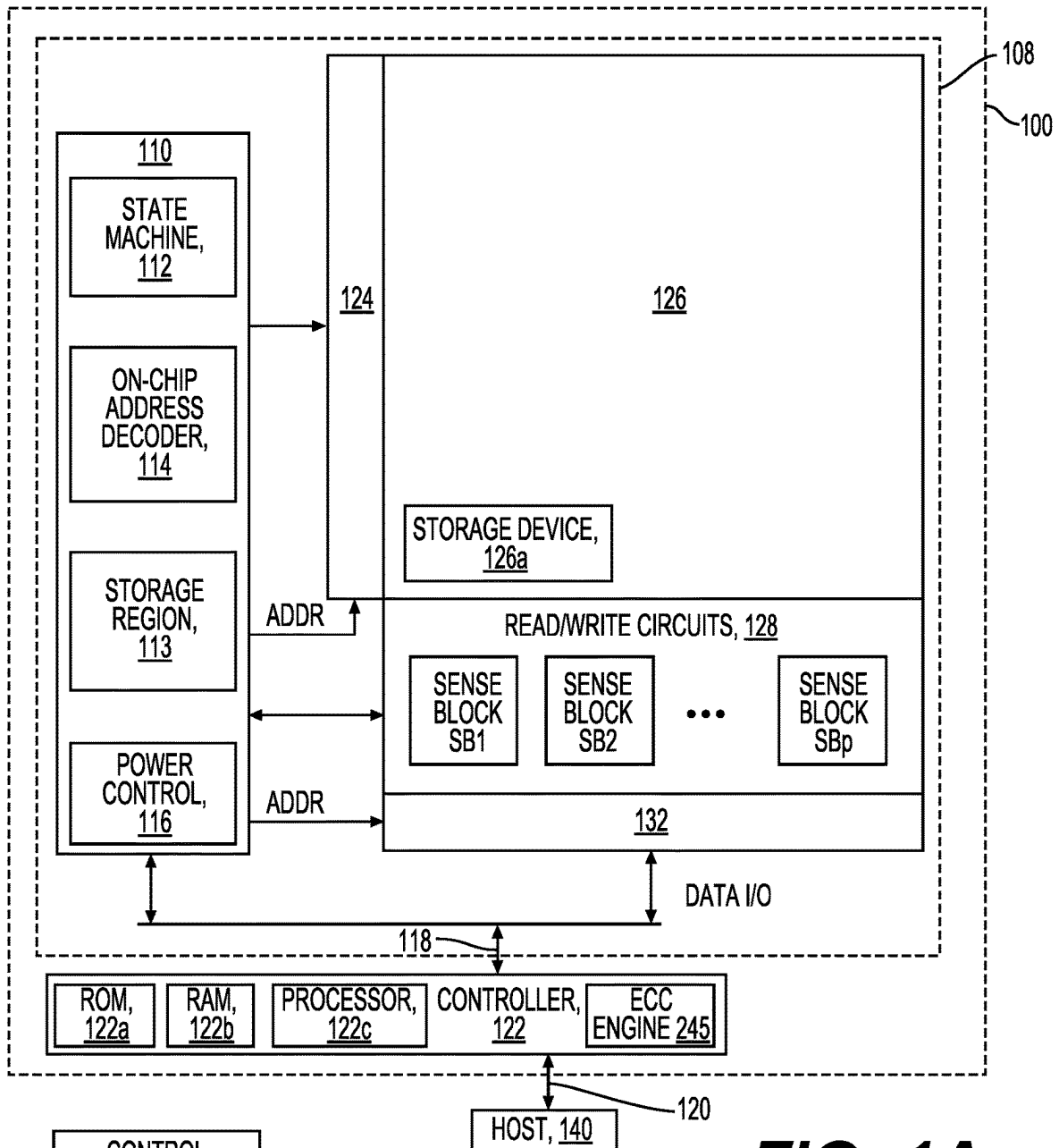


FIG. 1A

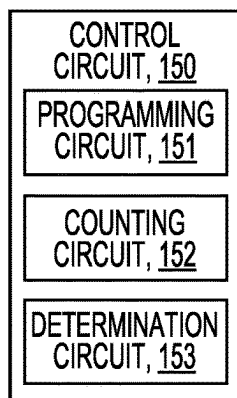


FIG. 1B

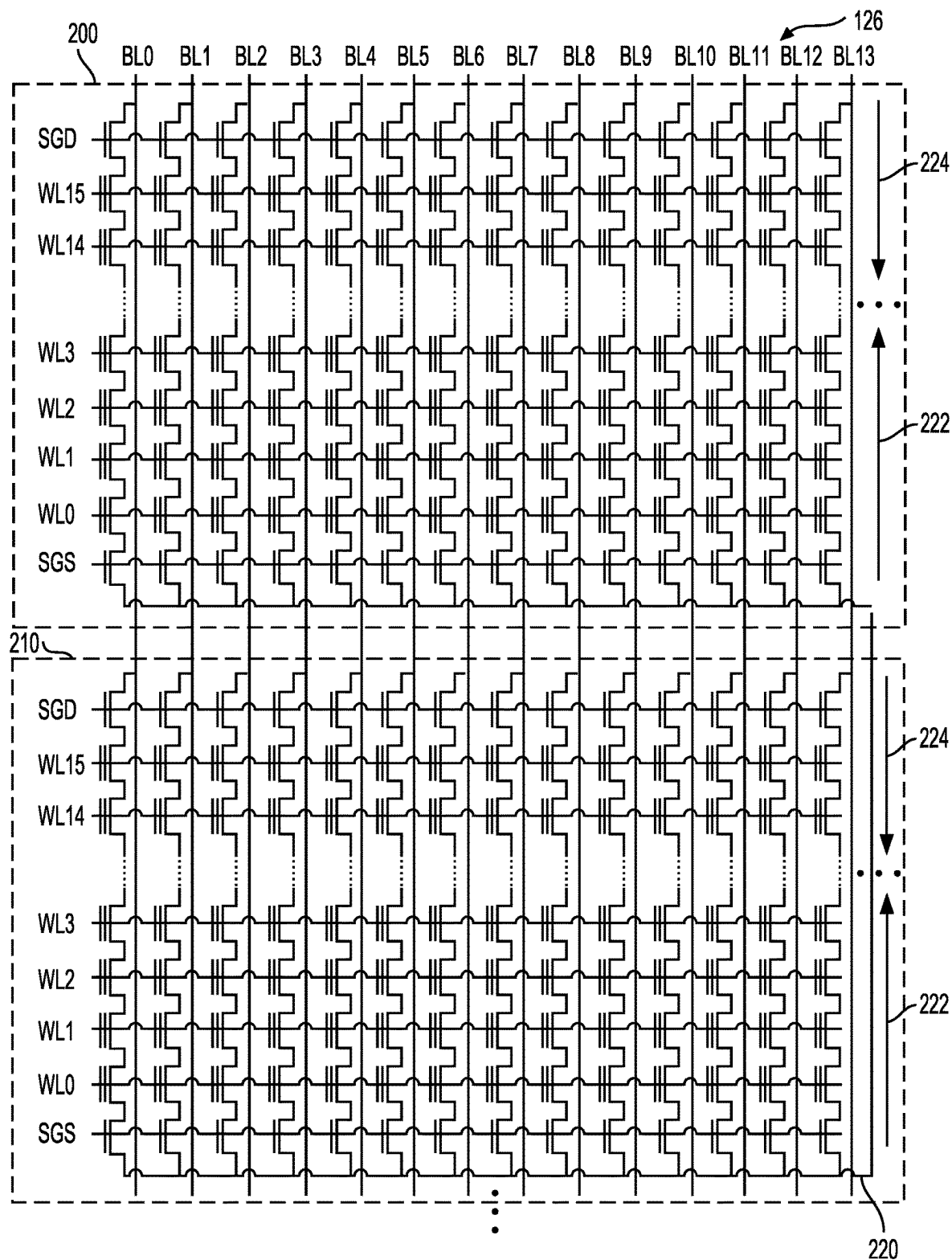


FIG. 2

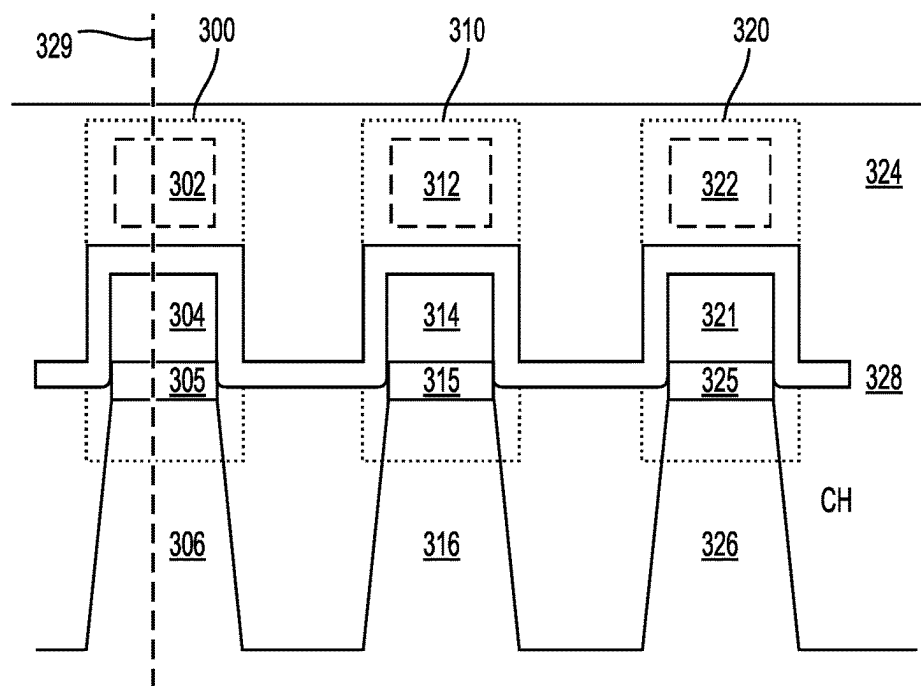


FIG. 3A

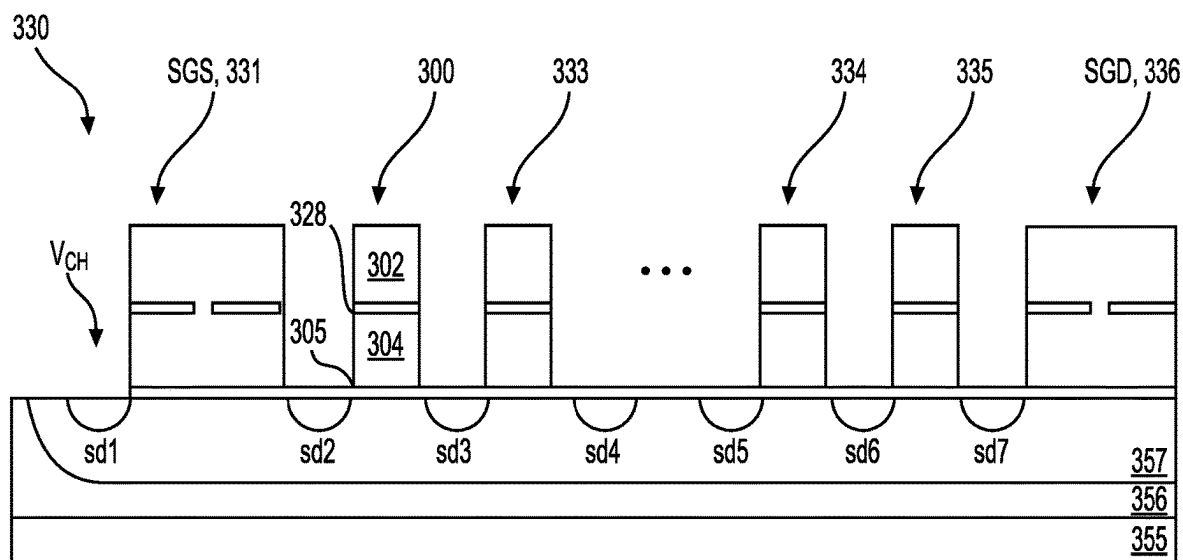


FIG. 3B

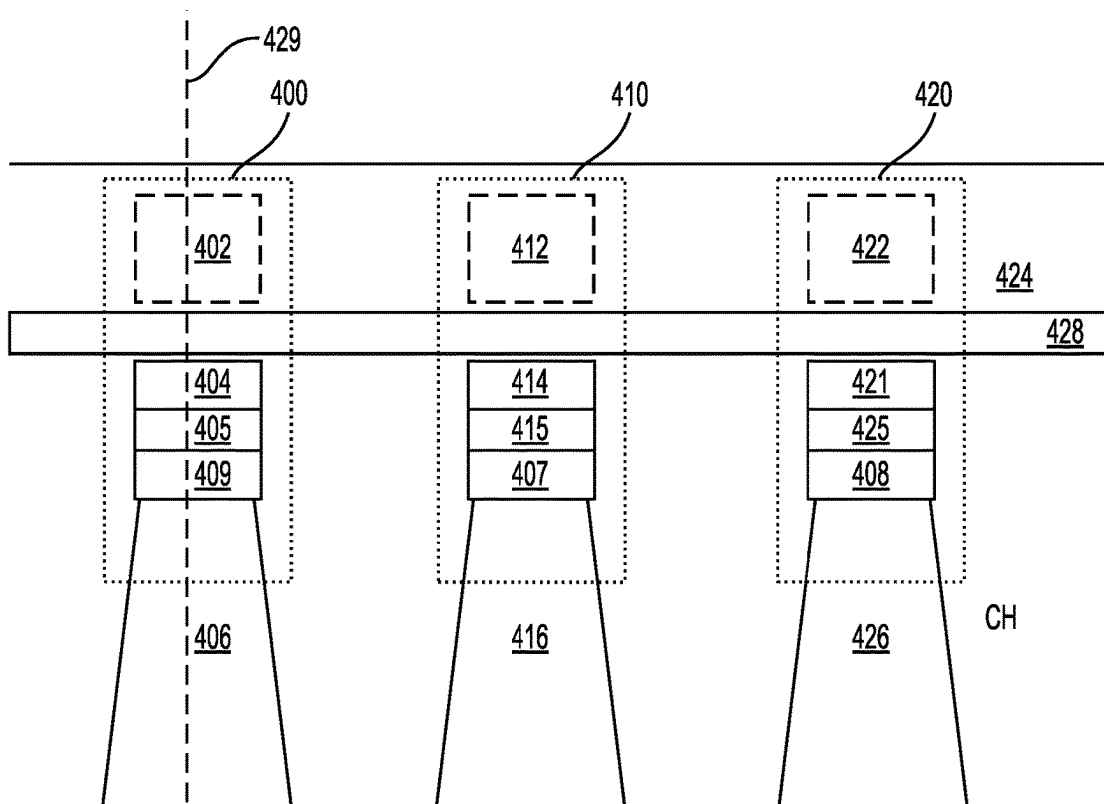


FIG. 4A

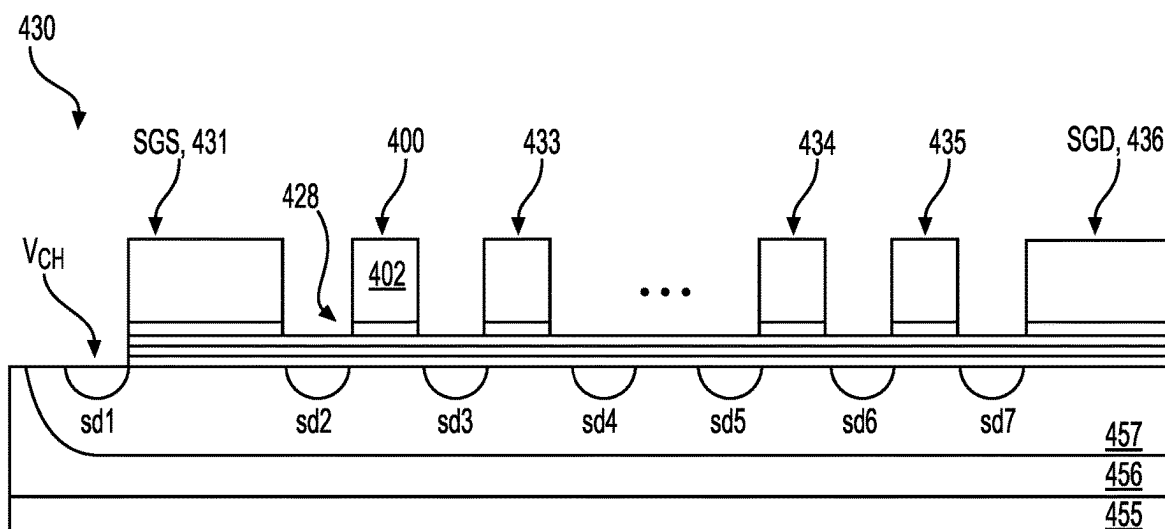


FIG. 4B

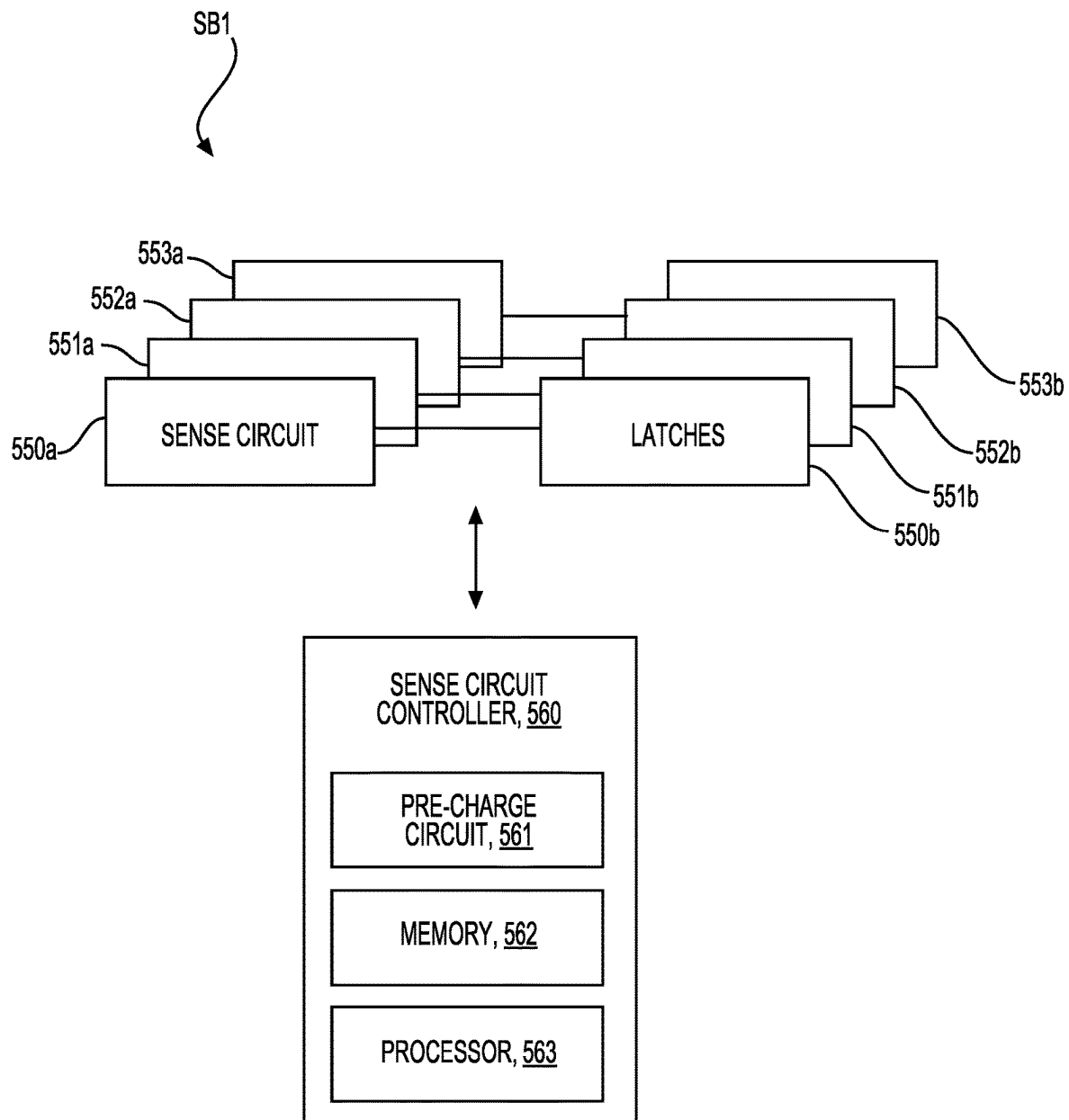
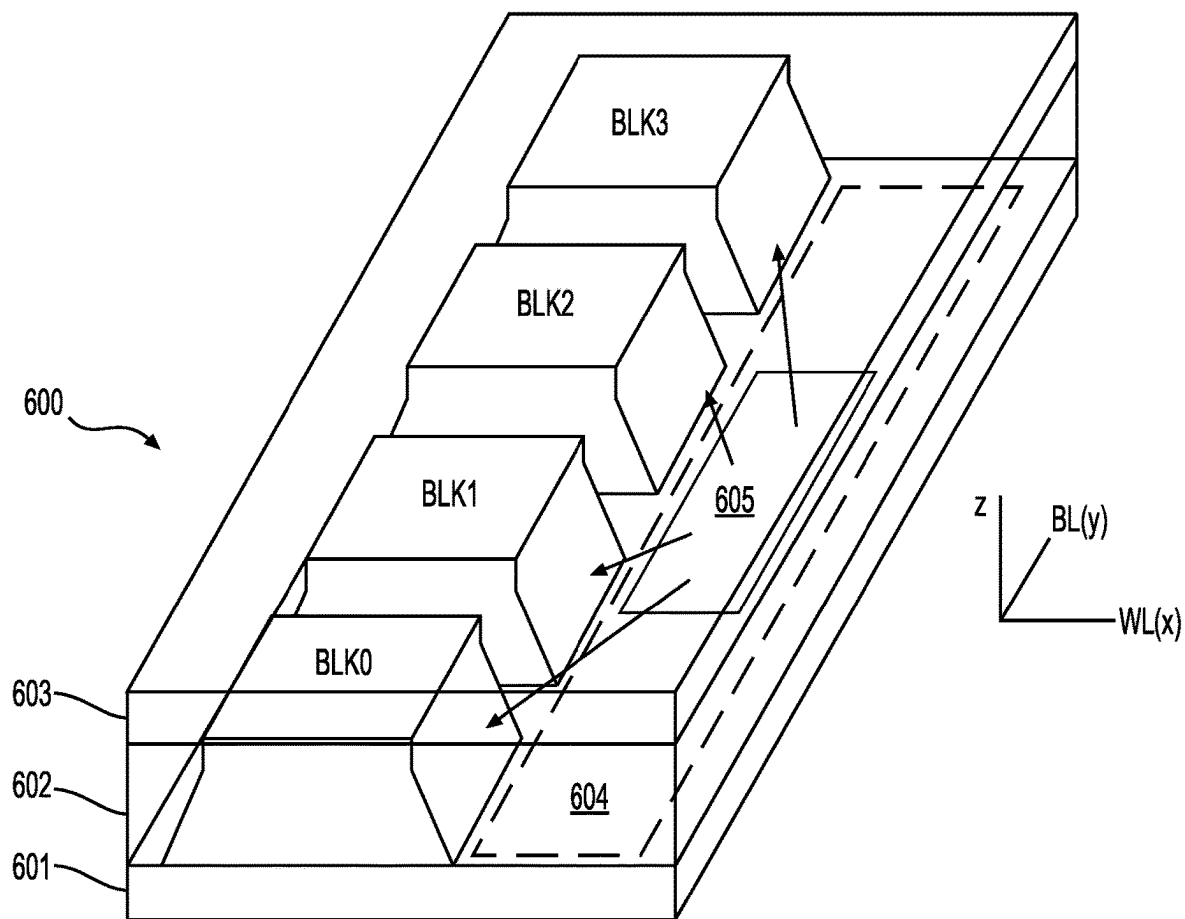


FIG. 5

**FIG. 6A**

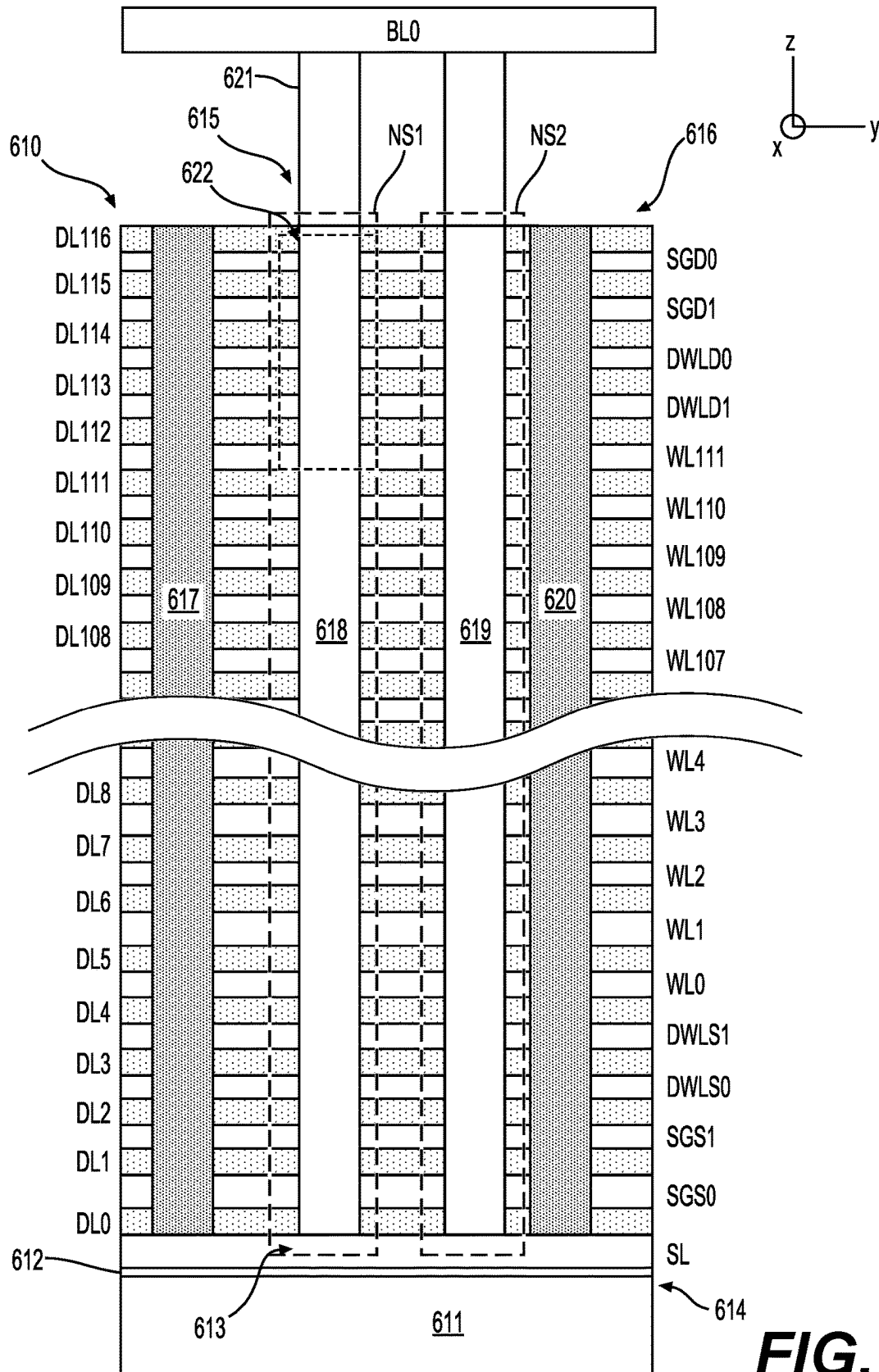


FIG. 6B

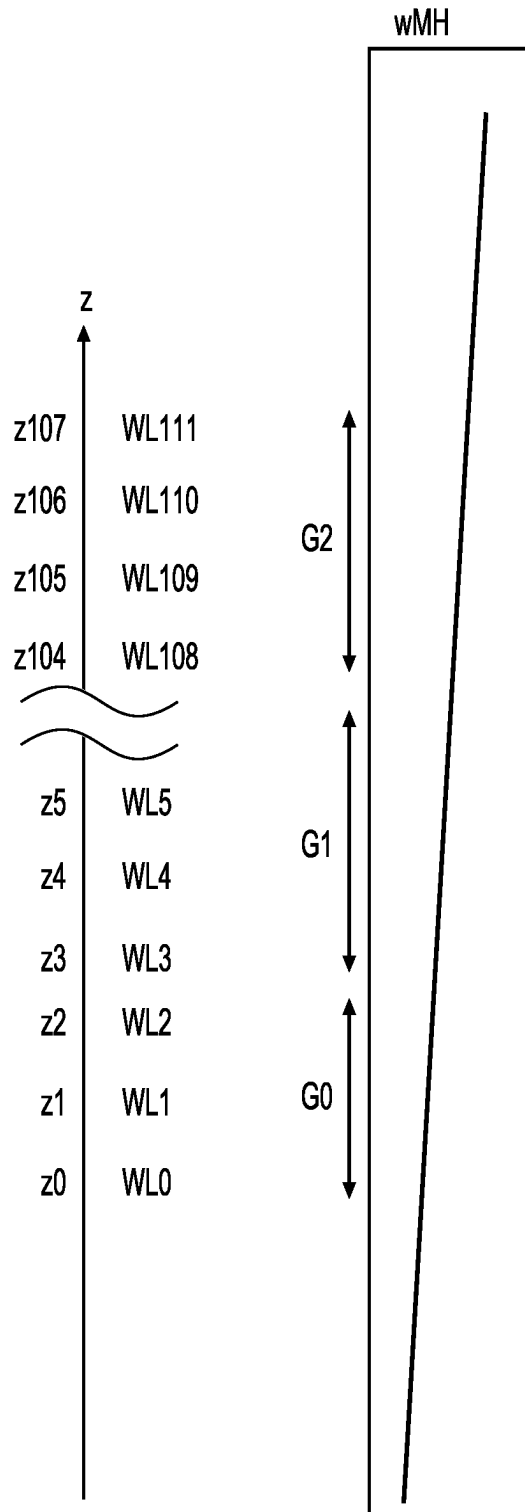


FIG. 6C

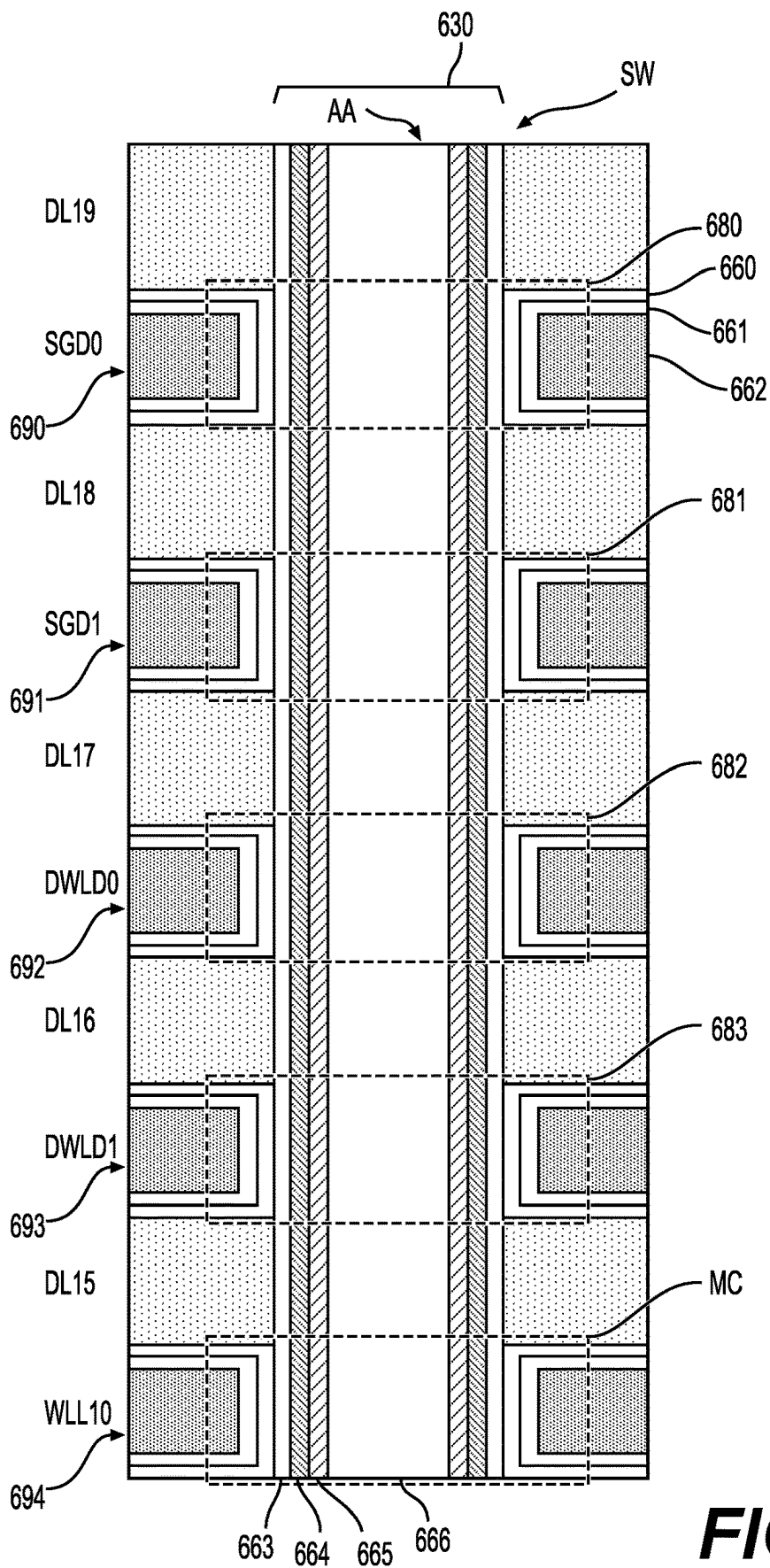


FIG. 6D

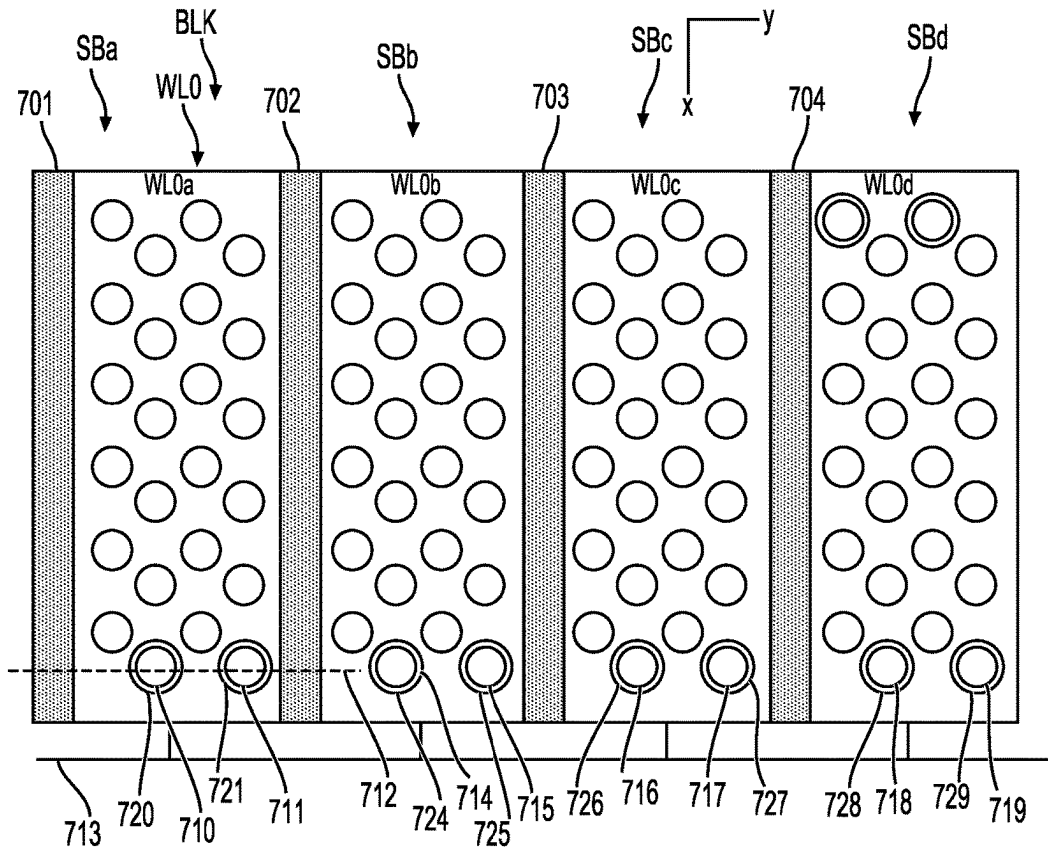


FIG. 7A

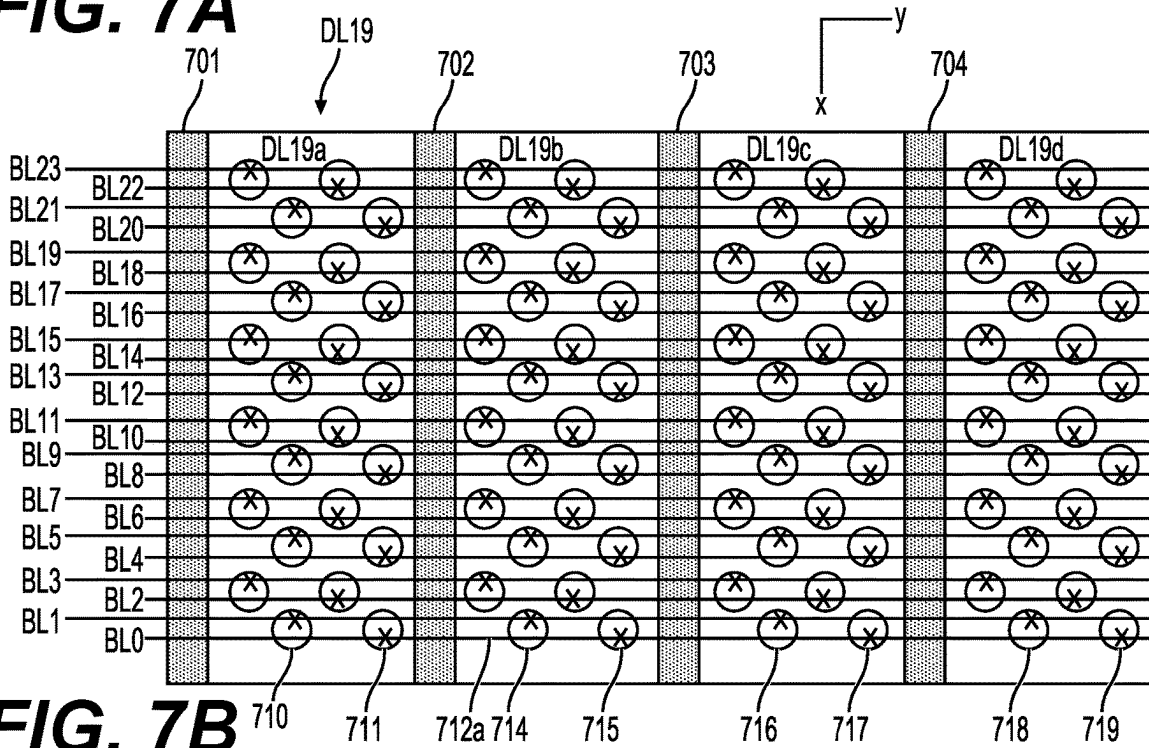
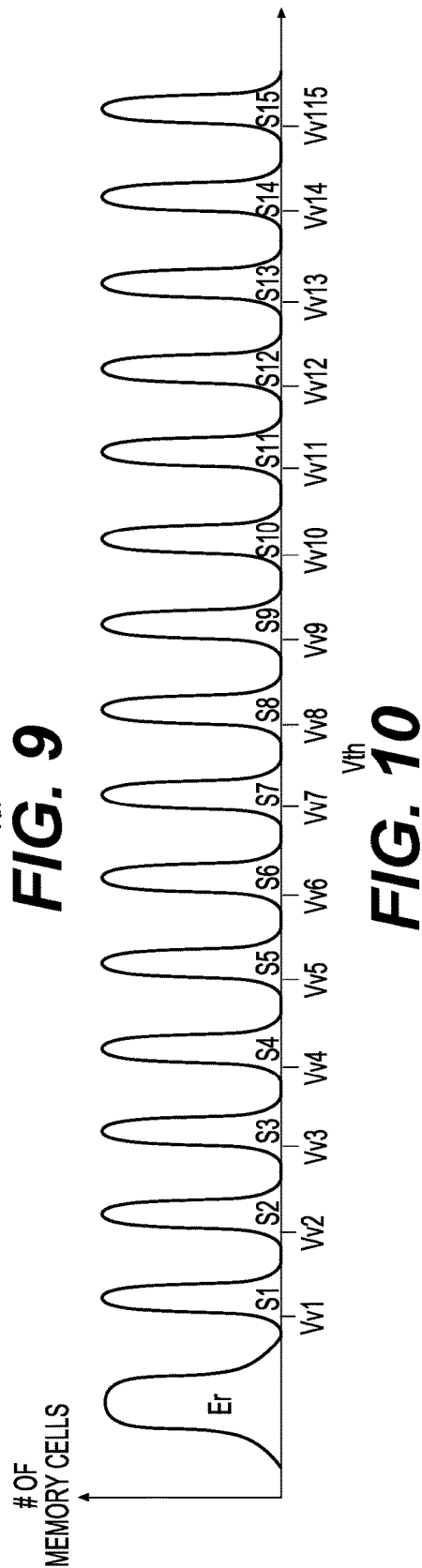
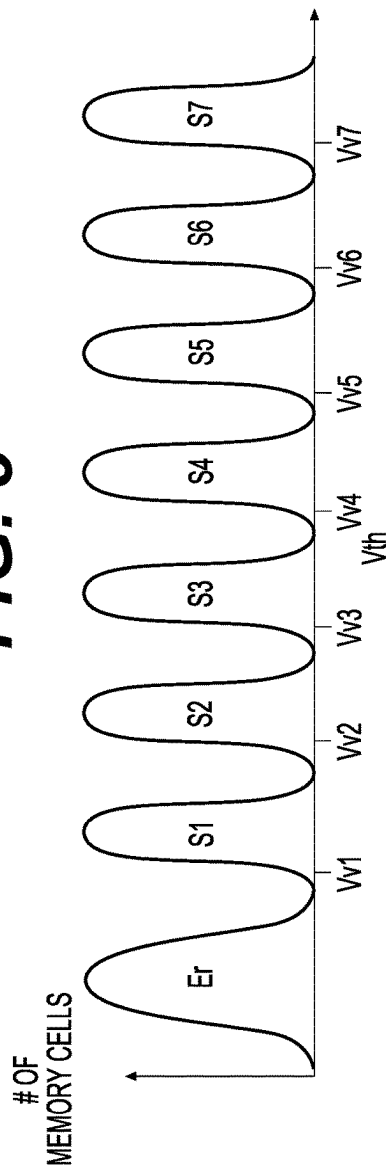
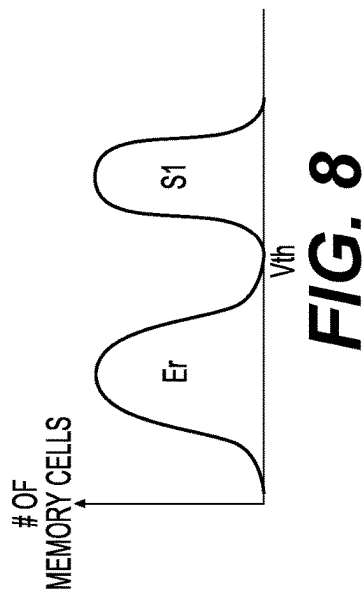


FIG. 7B



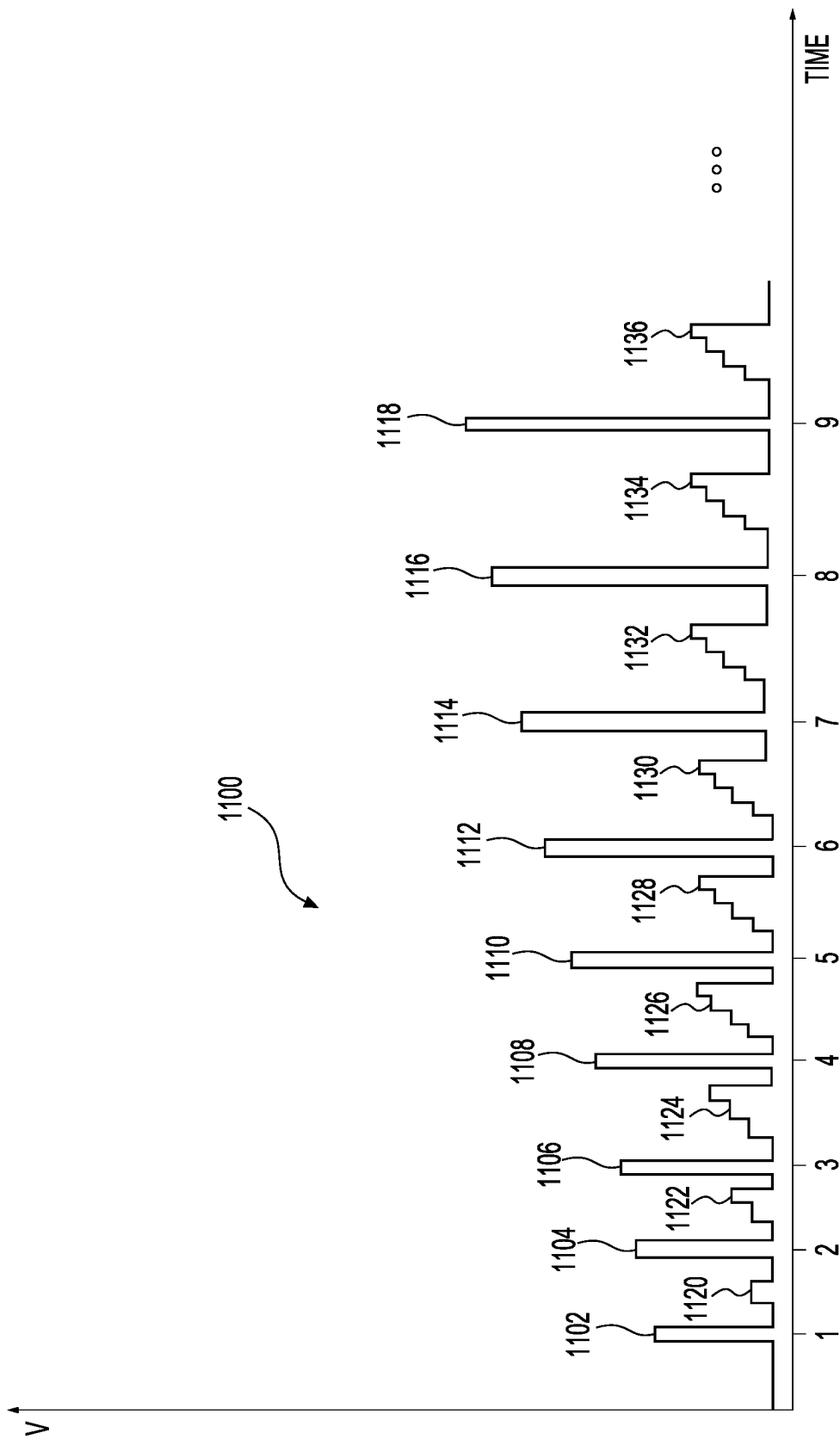


FIG. 11

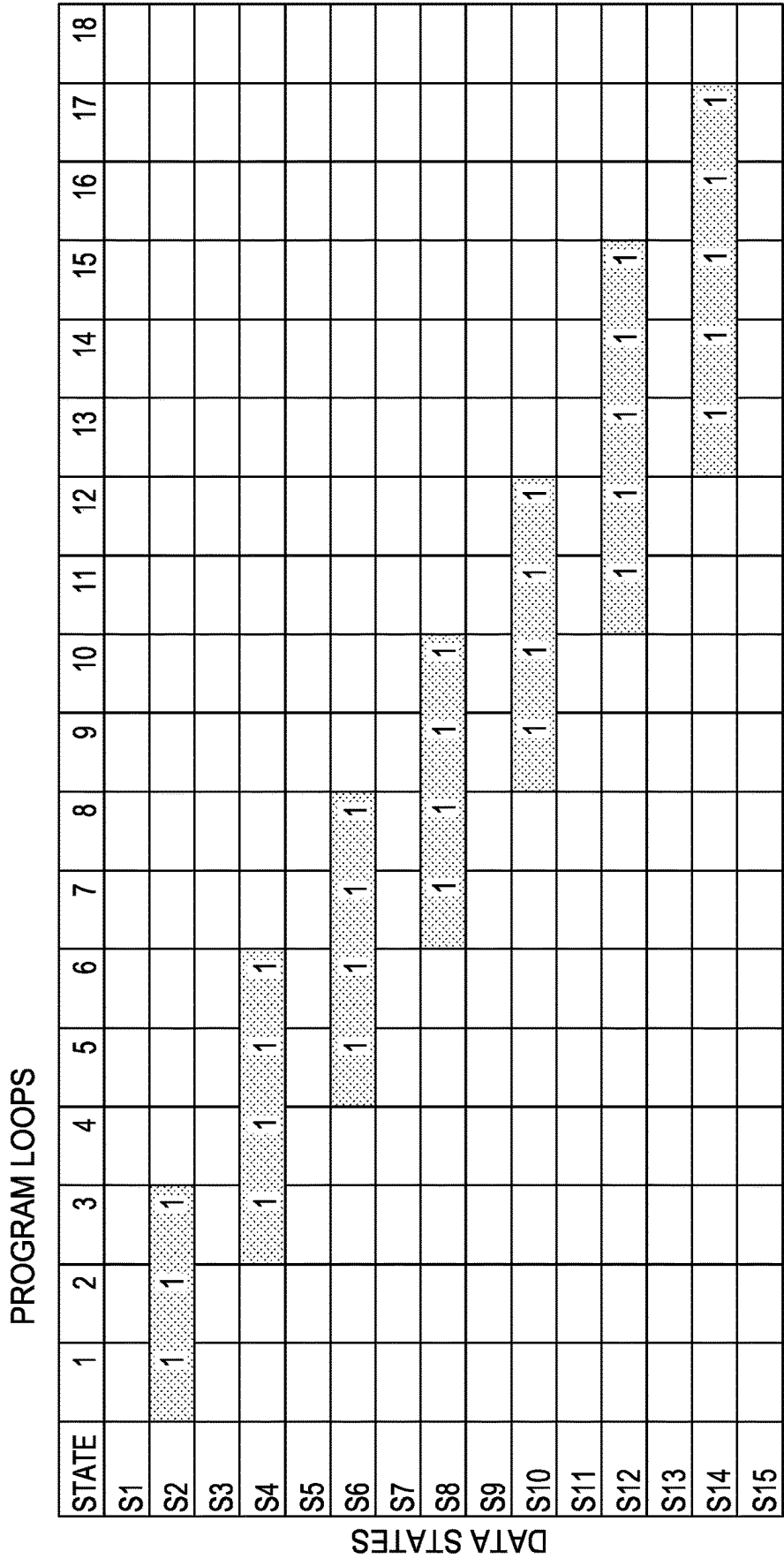


FIG. 12

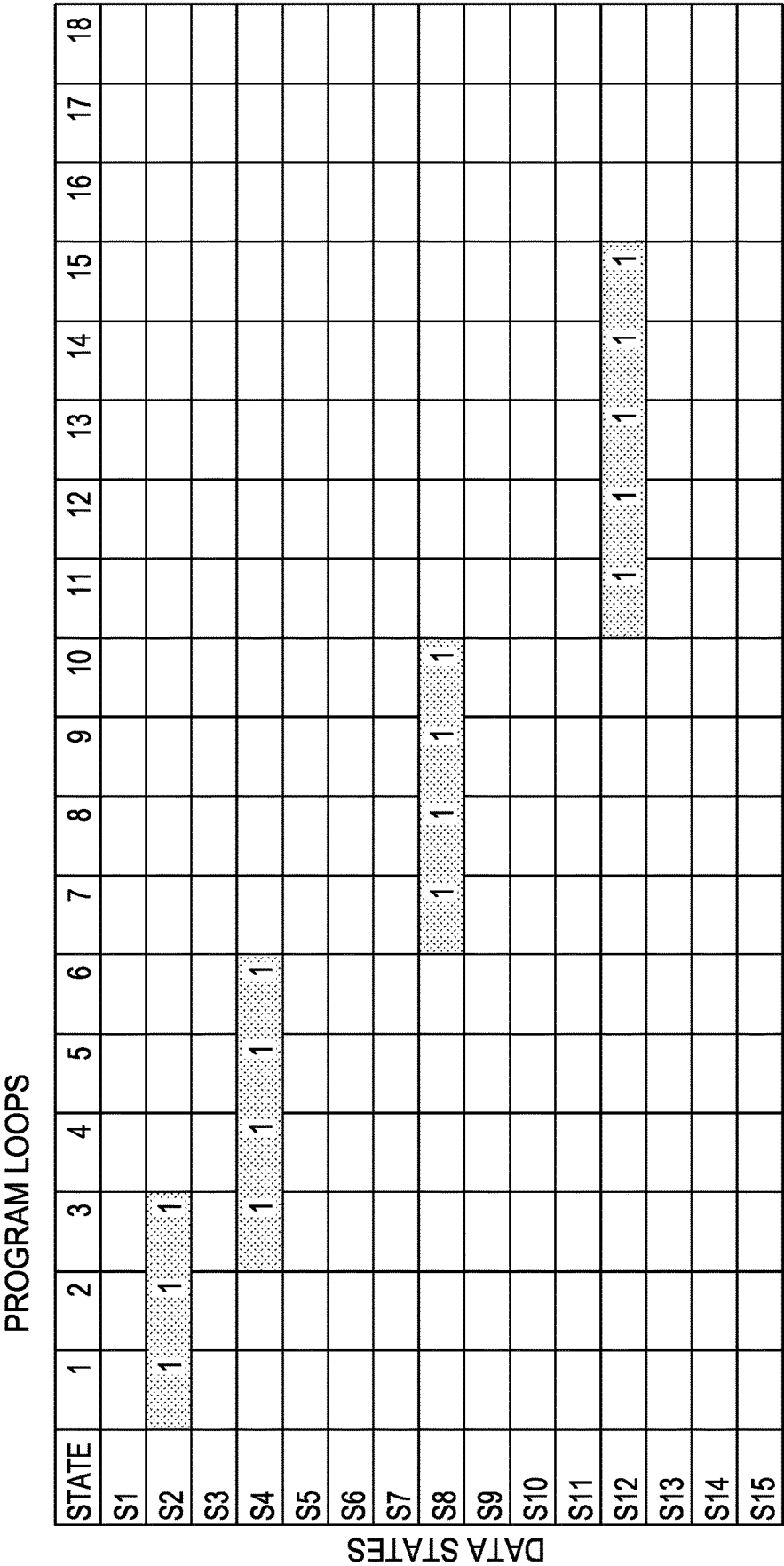


FIG. 13

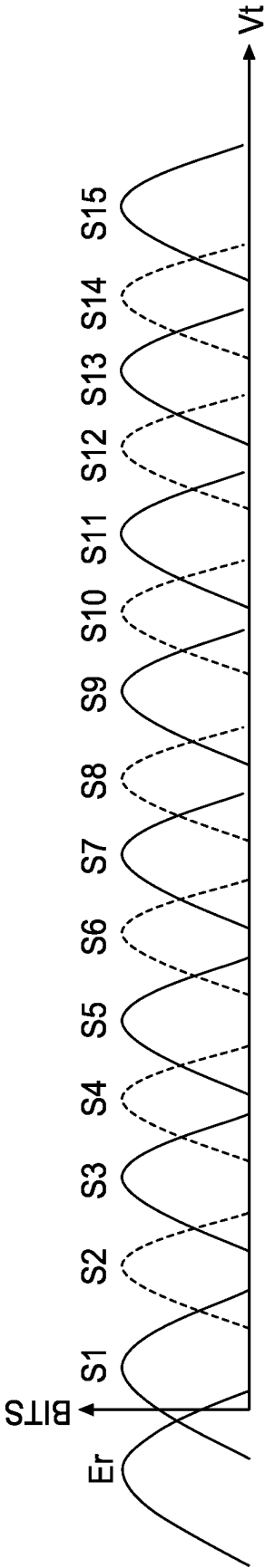


FIG. 14

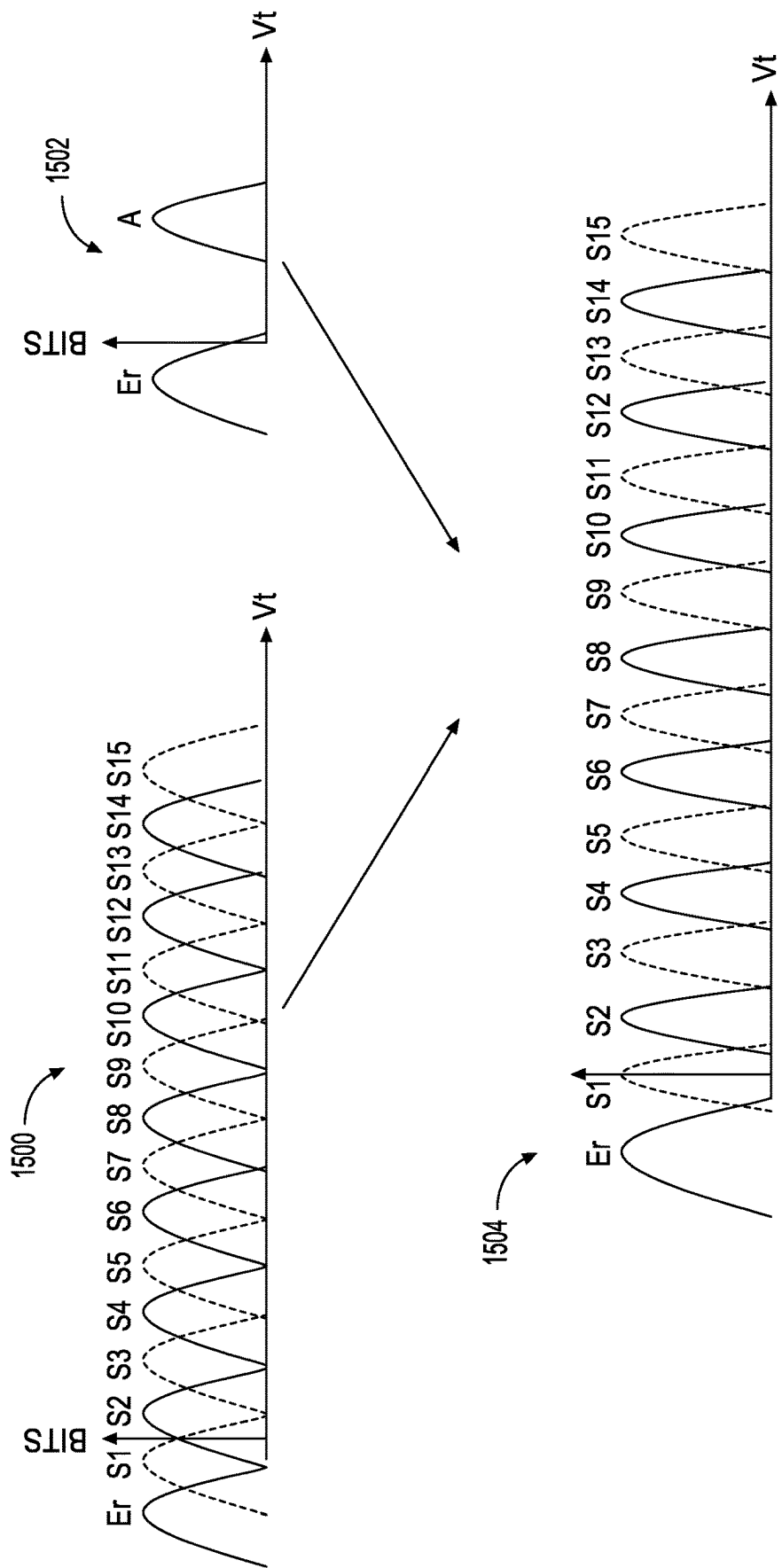


FIG. 15

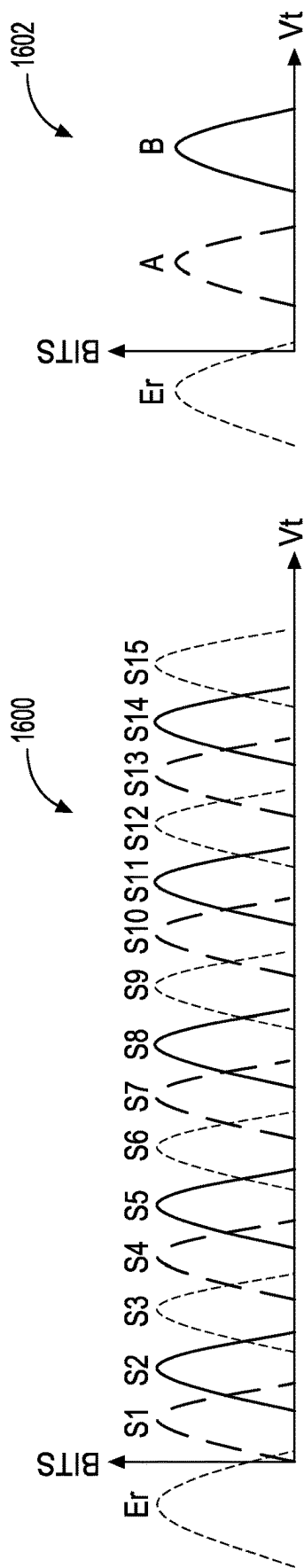


FIG. 16A

FIG. 16B

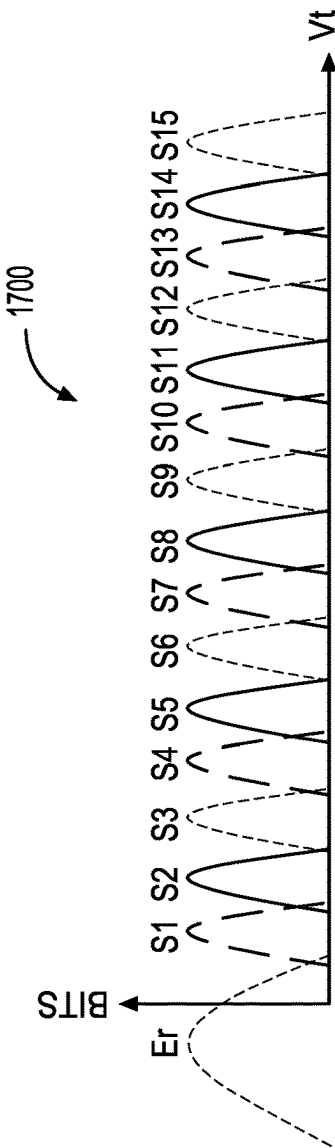
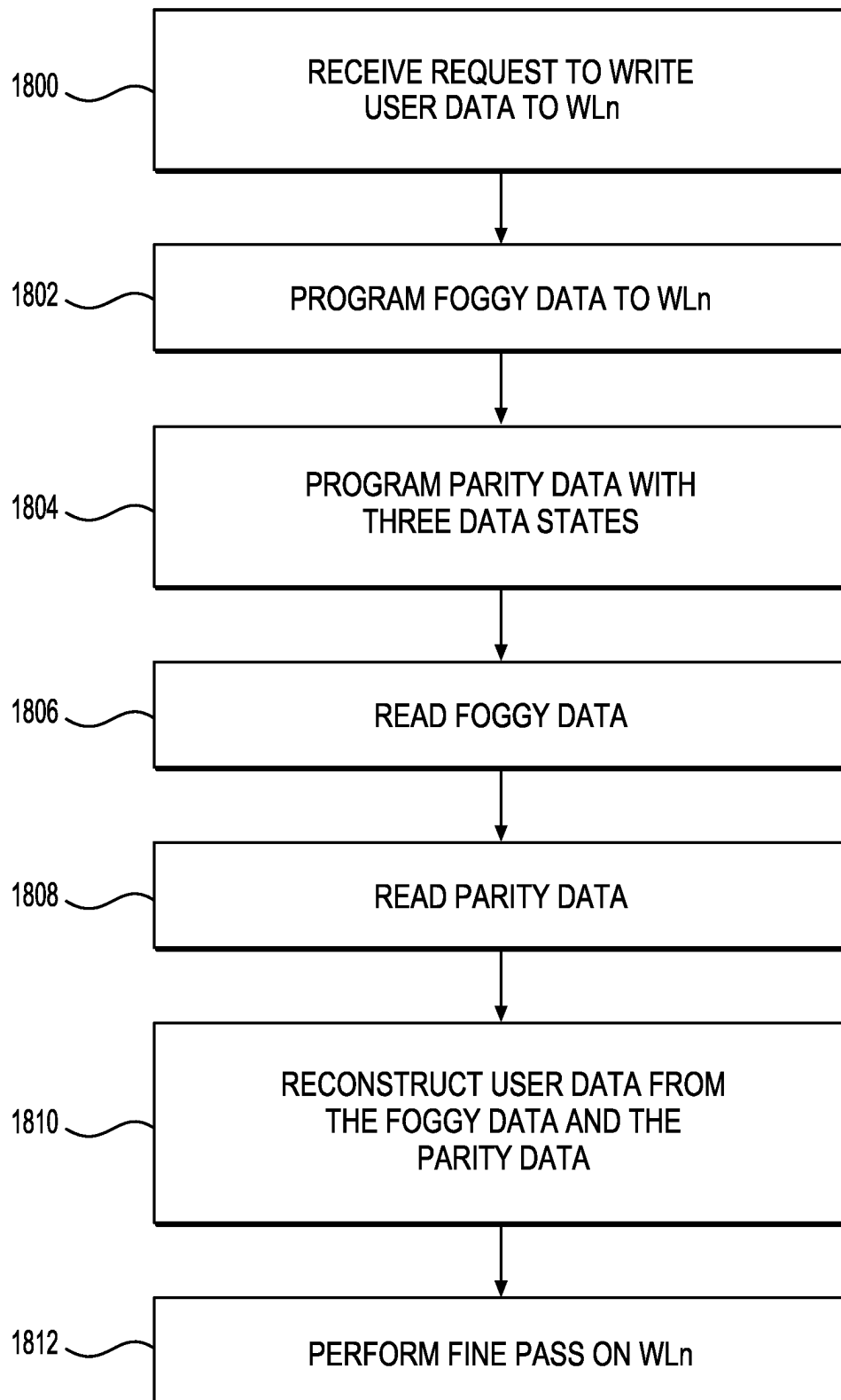


FIG. 17

**FIG. 18**

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MULTI-STAGE PROGRAMMING TECHNIQUES WITH THREE STATES PER MEMORY CELL PARITY

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Application No. 63/434,293, filed on Dec. 21, 2022. The entire disclosure of the application referenced above is incorporated herein by reference.

BACKGROUND

1. Field

The present disclosure is related generally to the operation of non-volatile memory and more particularly to improved multi-pass programming techniques to improve resource utilization.

2. Related Art

Semiconductor memory is widely used in various electronic devices, such as cellular telephones, digital cameras, personal digital assistants, medical electronics, mobile computing devices, servers, solid state drives, non-mobile computing devices and other devices. Semiconductor memory may comprise non-volatile memory or volatile memory. A non-volatile memory allows information to be stored and retained even when the non-volatile memory is not connected to a source of power, e.g., a battery.

NAND memory devices include a chip with a plurality of memory blocks, each of which includes an array of memory cells arranged in a plurality of word lines. Programming the memory cells of a word line to retain data typically occurs in a plurality of program loops, each of which includes the application of a programming pulse to a control gate of the word line and, optionally, a verify operation to sense the threshold voltages of the memory cells after the programming pulse is over.

In some programming techniques, programming a page of user data into the memory cells of a selected word line occurs in two programming passes: a first (foggy) pass and a second (fine) pass. In the foggy pass, the data is programmed into the memory cells at high performance (programming speed) but with low reliability. In the fine pass, the data is more accurately programmed into the memory cells. In some known techniques, the page of user data must be maintained somewhere in the memory device until the fine pass is completed due to the low reliability of the data in the selected word line following only the foggy pass. This maintenance of the page of user data between the foggy and fine passes consumes resources that could otherwise be dedicated to other operations.

SUMMARY

One aspect of the present disclosure is related to a method of programming a memory device. The method includes the step of preparing a memory block that includes an array of memory cells arranged in a plurality of word lines. The method also includes the step of programming final data into a selected word line of the plurality of word lines in a multi-pass programming operation. The multi-pass programming operation includes a first pass and a second pass. The first pass includes programming foggy data to the

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memory cells of the selected word line and programming parity data in the memory device. The parity data includes three possible data states. Prior to the second pass, the foggy data and the parity data are read, and the final data is reconstructed from the foggy data and the parity data. In the second pass, the memory cells of the selected word line are programmed from the foggy data to the final data.

According to another aspect of the present disclosure, the final data includes four bits per memory cell of the selected word line with each memory cell either remaining in an erased data state or being programmed to one of fifteen programmed data states.

According to yet another aspect of the present disclosure, the first pass includes a plurality of program loops. Each program loop includes a programming pulse and a verify operation. The verify operation is performed at a plurality of checkpoint data states. The plurality of checkpoint data states being less than the fifteen programmed data states.

According to still another aspect of the present disclosure, the plurality of checkpoint data states includes exactly five checkpoint data states.

According to a further aspect of the present disclosure, the plurality of checkpoint data states includes more than five checkpoint data states.

According to yet a further aspect of the present disclosure, the memory cells being programmed to non-checkpoint data states are verified at a previous one of the plurality of checkpoint data states and then are programmed in at least one program loop that includes a programming pulse and no verify operation.

According to still a further aspect of the present disclosure, the page of parity data is stored in NAND or DRAM.

Another aspect of the present direction is related to a memory device. The memory device includes a memory block with an array of memory cells arranged in a plurality of word lines. The memory device also includes control circuitry that is configured to program final data into a selected word line of the plurality of word lines in a multi-pass programming operation that includes a first pass and a second pass. In the first pass, the control circuitry is configured to program the memory cells of the selected word line to foggy data and program parity data in the memory device. The parity data includes three possible data states. Prior to the second pass, the control circuitry is configured to read the foggy data and the parity data and reconstruct the final data from the foggy data and the parity data. In the second pass, the control circuitry is configured to program the memory cells of the selected word line from the foggy data to the final data.

According to another aspect of the present disclosure, the final data includes four bits per memory cell of the selected word line with each memory cell either remaining in an erased data state or being programmed to one of fifteen programmed data states.

According to yet another aspect of the present disclosure, the first pass includes a plurality of program loops. In each program loop, the control circuitry is configured to apply a programming pulse and conduct a verify operation to the selected word line. The verify operation is performed by the control circuitry at a plurality of checkpoint data states. The plurality of checkpoint data states being less than the fifteen programmed data states.

According to still another aspect of the present disclosure, the plurality of checkpoint data states includes exactly five checkpoint data states.

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According to a further aspect of the present disclosure, the plurality of checkpoint data states includes more than five checkpoint data states.

According to yet a further aspect of the present disclosure, the control circuitry is configured to verify the memory cells being programmed to non-checkpoint data states at a previous one of the plurality of checkpoint data states and then program those memory cells being programmed to non-checkpoint data states in at least one program loop that includes a programming pulse and no verify operation.

According to still a further aspect of the present disclosure, the control circuitry is configured to store the page of parity data in NAND or DRAM.

Yet another aspect of the present disclosure is related to an apparatus. The apparatus includes a memory block that includes an array of memory cells arranged in a plurality of word lines. The apparatus also includes a programming means for programming final data into a selected word line of the plurality of word lines in a multi-pass programming operation that includes a first pass and a second pass. The programming means is configured to, in the first pass, program the memory cells of the selected word line to foggy data and program corresponding parity data in the memory device. The parity data includes three possible data states. Prior to the second pass, the control circuitry is configured to read the foggy data and the parity data and reconstruct the final data from the foggy data and the parity data. In the second pass, the control circuitry is configured to program the memory cells of the selected word line from the foggy data to the final data.

According to another aspect of the present disclosure, the final data includes four bits per memory cell of the selected word line with each memory cell either remaining in an erased data state or being programmed to one of fifteen programmed data states.

According to yet another aspect of the present disclosure, the first pass includes a plurality of program loops. In each program loop, the programming means is configured to apply a programming pulse and a verify operation to the selected word line. The verify operation is performed by the programming means at a plurality of checkpoint data states. The plurality of checkpoint data states is less than the fifteen programmed data states.

According to still another aspect of the present disclosure, the plurality of checkpoint data states includes exactly five checkpoint data states.

According to a further aspect of the present disclosure, the plurality of checkpoint data states includes more than five checkpoint data states.

According to yet a further aspect of the present disclosure, the programming means is configured to verify the memory cells being programmed to non-checkpoint data states at a previous one of the plurality of checkpoint data states and then program those memory cells being programmed to non-checkpoint data states in at least one program loop that includes a programming pulse and no verify operation.

BRIEF DESCRIPTION OF THE DRAWINGS

A more detailed description is set forth below with reference to example embodiments depicted in the appended figures. Understanding that these figures depict only example embodiments of the disclosure and are, therefore, not to be considered limiting of its scope. The disclosure is described and explained with added specificity and detail through the use of the accompanying drawings in which:

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FIG. 1A is a block diagram of an example memory device;

FIG. 1B is a block diagram of an example control circuit;

FIG. 2 depicts blocks of memory cells in an example two-dimensional configuration of the memory array of FIG. 1A;

FIG. 3A and FIG. 3B depict cross-sectional views of example floating gate memory cells in NAND strings;

FIG. 4A and FIG. 4B depict cross-sectional views of example charge-trapping memory cells in NAND strings;

FIG. 5 depicts an example block diagram of the sense block SB1 of FIG. 1;

FIG. 6A is a perspective view of a set of blocks in an example three-dimensional configuration of the memory array of FIG. 1;

FIG. 6B depicts an example cross-sectional view of a portion of one of the blocks of FIG. 6A;

FIG. 6C depicts a plot of memory hole diameter in the stack of FIG. 6B;

FIG. 6D depicts a close-up view of region 622 of the stack of FIG. 6B;

FIG. 7A depicts a top view of an example word line layer WL0 of the stack of FIG. 6B;

FIG. 7B depicts a top view of an example top dielectric layer DL116 of the stack of FIG. 6B;

FIG. 8 depicts a threshold voltage distribution of a page of memory cells programmed to one bit per memory cell (SLC);

FIG. 9 depicts a threshold voltage distribution of a page of memory cells programmed to three bits per memory cell (TLC);

FIG. 10 depicts a threshold voltage distribution of a page of memory cells programmed to four bits per memory cell (QLC);

FIG. 11 depicts a voltage waveform of the voltage applied to a control gate of a selected word line during an example programming operation;

FIG. 12 is a chart depicting which data states are verified at which program loops in one type of reduced verify programming technique;

FIG. 13 is a chart depicting which data states are verified at which program loops in another type of reduced verify programming technique;

FIG. 14 is a threshold voltage distribution plot of a page of foggy data following a seven checkpoint reduced verify foggy pass;

FIG. 15 illustrates the threshold voltage distribution charts of a page of foggy data and a page of parity data being used to reconstruct a page of final data;

FIG. 16A illustrates a threshold voltage distribution chart of a page of foggy data programmed using a five checkpoint reduced verify operation;

FIG. 16B illustrates a threshold voltage distribution chart of a page of parity data;

FIG. 17 illustrates a threshold voltage distribution chart of a page of final data following reconstruction of the final data and a fine programming pass; and

FIG. 18 is a flow chart depicting the steps of programming a selected word line according to an example embodiment of the present disclosure.

DESCRIPTION OF THE ENABLING EMBODIMENTS

The present disclosure is related to a multi-pass programming technique which reduces recourse utilization in a flash memory device when programming a page of user data

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(sometimes hereinafter referred to as “final data”) into a selected word line. In a first pass, the memory cells of a selected word line are programmed at high speed to contain a page of foggy data. Also during the first pass, a page of parity data is programmed elsewhere in the memory device (e.g., in NAND or DRAM). Prior to the second pass, the page of foggy data and the page of parity data are both read by control circuitry in the memory device, and then the page of final data is reconstructed using the foggy data and the parity data. In the second pass, the memory cells of the selected word line are programmed from the page of foggy data to the page of final data without erase. These techniques allow for more efficient resource utilization in the memory device by eliminating the need to maintain the page of final data between the first and second passes. According to the present disclosure, each memory cell used for the parity data storage contains exactly three possible data states. By providing the parity data with three possible data states, the reliability of the final data reconstruction is improved with little or no loss in performance (programming speed). These techniques can also be used in conjunction with reduced verify techniques where fewer than all data states are verified during programming. These and other features of the programming techniques are discussed in further detail below.

FIG. 1A is a block diagram of an example memory device **100** is configured to operate according to the programming techniques of the present disclosure. The memory die **108** includes a memory structure **126** of memory cells, such as an array of memory cells, control circuitry **110**, and read/write circuits **128**. The memory structure **126** is addressable by word lines via a row decoder **124** and by bit lines via a column decoder **132**. The read/write circuits **128** include multiple sense blocks SB1, SB2, . . . SBp (sensing circuitry) and allow a page of memory cells to be read or programmed in parallel. Typically, a controller **122** is included in the same memory device **100** (e.g., a removable storage card) as the one or more memory die **108**. Commands and data are transferred between the host **140** and controller **122** via a data bus **120**, and between the controller and the one or more memory die **108** via lines **118**.

The memory structure **126** can be two-dimensional or three-dimensional. The memory structure **126** may comprise one or more array of memory cells including a three-dimensional array. The memory structure **126** may comprise a monolithic three-dimensional memory structure in which multiple memory levels are formed above (and not in) a single substrate, such as a wafer, with no intervening substrates. The memory structure **126** may comprise any type of non-volatile memory that is monolithically formed in one or more physical levels of arrays of memory cells having an active area disposed above a silicon substrate. The memory structure **126** may be in a non-volatile memory device having circuitry associated with the operation of the memory cells, whether the associated circuitry is above or within the substrate.

The control circuitry **110** cooperates with the read/write circuits **128** to perform memory operations on the memory structure **126**, and includes a state machine **112**, an on-chip address decoder **114**, and a power control module **116**. The state machine **112** provides chip-level control of memory operations.

A storage region **113** may, for example, be provided for programming parameters. The programming parameters may include a program voltage, a program voltage bias, position parameters indicating positions of memory cells, contact line connector thickness parameters, a verify volt-

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age, and/or the like. The position parameters may indicate a position of a memory cell within the entire array of NAND strings, a position of a memory cell as being within a particular NAND string group, a position of a memory cell on a particular plane, and/or the like. The contact line connector thickness parameters may indicate a thickness of a contact line connector, a substrate or material that the contact line connector is comprised of, and/or the like.

The on-chip address decoder **114** provides an address interface between the host or a memory controller to the hardware address used by the decoders **124** and **132**. The power control module **116** controls the power and voltages supplied to the word lines and bit lines during memory operations. It can include drivers for word lines, SGS and SGD transistors, and source lines. The sense blocks can include bit line drivers, in one approach. An SGS transistor is a select gate transistor at a source end of a NAND string, and an SGD transistor is a select gate transistor at a drain end of a NAND string.

In some embodiments, some of the components can be combined. In various designs, one or more of the components (alone or in combination), other than memory structure **126**, can be thought of as at least one control circuit which is configured to perform the actions described herein. For example, a control circuit may include any one of, or a combination of, control circuitry **110**, state machine **112**, decoders **114/132**, power control module **116**, sense blocks SBb, SB2, . . . , SBp, read/write circuits **128**, controller **122**, and so forth.

The control circuits can include a programming circuit configured to perform a program and verify operation for one set of memory cells, wherein the one set of memory cells comprises memory cells assigned to represent one data state among a plurality of data states and memory cells assigned to represent another data state among the plurality of data states; the program and verify operation comprising a plurality of program and verify iterations; and in each program and verify iteration, the programming circuit performs programming for the one selected word line after which the programming circuit applies a verification signal to the selected word line. The control circuits can also include a counting circuit configured to obtain a count of memory cells which pass a verify test for the one data state. The control circuits can also include a determination circuit configured to determine, based on an amount by which the count exceeds a threshold, if a programming operation is completed.

For example, FIG. 1B is a block diagram of an example control circuit **150** which comprises a programming circuit **151**, a counting circuit **152**, and a determination circuit **153**.

The off-chip controller **122** may comprise a processor **122c**, storage devices (memory) such as ROM **122a** and RAM **122b** and an error-correction code (ECC) engine **245**. The ECC engine can correct a number of read errors which are caused when the upper tail of a Vth distribution becomes too high. However, uncorrectable errors may exist in some cases. The techniques provided herein reduce the likelihood of uncorrectable errors.

The storage device(s) **122a**, **122b** comprise, code such as a set of instructions, and the processor **122c** is operable to execute the set of instructions to provide the functionality described herein. Alternately or additionally, the processor **122c** can access code from a storage device **126a** of the memory structure **126**, such as a reserved area of memory cells in one or more word lines. For example, code can be used by the controller **122** to access the memory structure **126** such as for programming, read and erase operations. The

code can include boot code and control code (e.g., set of instructions). The boot code is software that initializes the controller 122 during a booting or startup process and enables the controller 122 to access the memory structure 126. The code can be used by the controller 122 to control one or more memory structures 126. Upon being powered up, the processor 122c fetches the boot code from the ROM 122a or storage device 126a for execution, and the boot code initializes the system components and loads the control code into the RAM 122b. Once the control code is loaded into the RAM 122b, it is executed by the processor 122c. The control code includes drivers to perform basic tasks such as controlling and allocating memory, prioritizing the processing of instructions, and controlling input and output ports.

Generally, the control code can include instructions to perform the functions described herein including the steps of the flowcharts discussed further below and provide the voltage waveforms including those discussed further below.

In one embodiment, the host is a computing device (e.g., laptop, desktop, smartphone, tablet, digital camera) that includes one or more processors, one or more processor readable storage devices (RAM, ROM, flash memory, hard disk drive, solid state memory) that store processor readable code (e.g., software) for programming the one or more processors to perform the methods described herein. The host may also include additional system memory, one or more input/output interfaces and/or one or more input/output devices in communication with the one or more processors.

Other types of non-volatile memory in addition to NAND flash memory can also be used.

Semiconductor memory devices include volatile memory devices, such as dynamic random access memory ("DRAM") or static random access memory ("SRAM") devices, non-volatile memory devices, such as resistive random access memory ("ReRAM"), electrically erasable programmable read only memory ("EEPROM"), flash memory (which can also be considered a subset of EEPROM), ferroelectric random access memory ("FRAM"), and magnetoresistive random access memory ("MRAM"), and other semiconductor elements capable of storing information. Each type of memory device may have different configurations. For example, flash memory devices may be configured in a NAND or a NOR configuration.

The memory devices can be formed from passive and/or active elements, in any combinations. By way of non-limiting example, passive semiconductor memory elements include ReRAM device elements, which in some embodiments include a resistivity switching storage element, such as an anti-fuse or phase change material, and optionally a steering element, such as a diode or transistor. Further by way of non-limiting example, active semiconductor memory elements include EEPROM and flash memory device elements, which in some embodiments include elements containing a charge storage region, such as a floating gate, conductive nanoparticles, or a charge storage dielectric material.

Multiple memory elements may be configured so that they are connected in series or so that each element is individually accessible. By way of non-limiting example, flash memory devices in a NAND configuration (NAND memory) typically contain memory elements connected in series. A NAND string is an example of a set of series-connected transistors comprising memory cells and SG transistors.

A NAND memory array may be configured so that the array is composed of multiple memory strings in which a string is composed of multiple memory elements sharing a

single bit line and accessed as a group. Alternatively, memory elements may be configured so that each element is individually accessible, e.g., a NOR memory array. NAND and NOR memory configurations are examples, and memory elements may be otherwise configured. The semiconductor memory elements located within and/or over a substrate may be arranged in two or three dimensions, such as a two-dimensional memory structure or a three-dimensional memory structure.

In a two-dimensional memory structure, the semiconductor memory elements are arranged in a single plane or a single memory device level. Typically, in a two-dimensional memory structure, memory elements are arranged in a plane (e.g., in an x-y direction plane) which extends substantially parallel to a major surface of a substrate that supports the memory elements. The substrate may be a wafer over or in which the layer of the memory elements is formed or it may be a carrier substrate which is attached to the memory elements after they are formed. As a non-limiting example, the substrate may include a semiconductor such as silicon.

The memory elements may be arranged in the single memory device level in an ordered array, such as in a plurality of rows and/or columns. However, the memory elements may be arrayed in non-regular or non-orthogonal configurations. The memory elements may each have two or more electrodes or contact lines, such as bit lines and word lines.

A three-dimensional memory array is arranged so that memory elements occupy multiple planes or multiple memory device levels, thereby forming a structure in three dimensions (i.e., in the x, y and z directions, where the z-direction is substantially perpendicular and the x- and y-directions are substantially parallel to the major surface of the substrate).

As a non-limiting example, a three-dimensional memory structure may be vertically arranged as a stack of multiple two-dimensional memory device levels. As another non-limiting example, a three-dimensional memory array may be arranged as multiple vertical columns (e.g., columns extending substantially perpendicular to the major surface of the substrate, i.e., in the y direction) with each column having multiple memory elements. The columns may be arranged in a two-dimensional configuration, e.g., in an x-y plane, resulting in a three-dimensional arrangement of memory elements with elements on multiple vertically stacked memory planes. Other configurations of memory elements in three dimensions can also constitute a three-dimensional memory array.

By way of non-limiting example, in a three-dimensional array of NAND strings, the memory elements may be coupled together to form a NAND string within a single horizontal (e.g., x-y) memory device level. Alternatively, the memory elements may be coupled together to form a vertical NAND string that traverses across multiple horizontal memory device levels. Other three-dimensional configurations can be envisioned wherein some NAND strings contain memory elements in a single memory level while other strings contain memory elements which span through multiple memory levels. Three-dimensional memory arrays may also be designed in a NOR configuration and in a ReRAM configuration.

Typically, in a monolithic three-dimensional memory array, one or more memory device levels are formed above a single substrate. Optionally, the monolithic three-dimensional memory array may also have one or more memory layers at least partially within the single substrate. As a non-limiting example, the substrate may include a semicon-

ductor such as silicon. In a monolithic three-dimensional array, the layers constituting each memory device level of the array are typically formed on the layers of the underlying memory device levels of the array. However, layers of adjacent memory device levels of a monolithic three-dimensional memory array may be shared or have intervening layers between memory device levels.

Then again, two-dimensional arrays may be formed separately and then packaged together to form a non-monolithic memory device having multiple layers of memory. For example, non-monolithic stacked memories can be constructed by forming memory levels on separate substrates and then stacking the memory levels atop each other. The substrates may be thinned or removed from the memory device levels before stacking, but as the memory device levels are initially formed over separate substrates, the resulting memory arrays are not monolithic three-dimensional memory arrays. Further, multiple two-dimensional memory arrays or three-dimensional memory arrays (monolithic or non-monolithic) may be formed on separate chips and then packaged together to form a stacked-chip memory device.

FIG. 2 illustrates blocks **200**, **210** of memory cells in an example two-dimensional configuration of the memory array **126** of FIG. 1. The memory array **126** can include many such blocks **200**, **210**. Each example block **200**, **210** includes a number of NAND strings and respective bit lines, e.g., BL0, BL1, . . . which are shared among the blocks. Each NAND string is connected at one end to a drain-side select gate (SGD), and the control gates of the drain select gates are connected via a common SGD line. The NAND strings are connected at their other end to a source-side select gate (SGS) which, in turn, is connected to a common source line **220**. One hundred and twelve word lines, for example, WL0-WL111, extend between the SGSs and the SGDs. In some embodiments, the memory block may include more or fewer than one hundred and twelve word lines. For example, in some embodiments, a memory block includes one hundred and sixty-four word lines. In some cases, dummy word lines, which contain no user data, can also be used in the memory array adjacent to the select gate transistors. Such dummy word lines can shield the edge data word line from certain edge effects.

One type of non-volatile memory which may be provided in the memory array is a floating gate memory, such as of the type shown in FIGS. 3A and 3B. However, other types of non-volatile memory can also be used. As discussed in further detail below, in another example shown in FIGS. 4A and 4B, a charge-trapping memory cell uses a non-conductive dielectric material in place of a conductive floating gate to store charge in a non-volatile manner. A triple layer dielectric formed of silicon oxide, silicon nitride and silicon oxide ("ONO") is sandwiched between a conductive control gate and a surface of a semi-conductive substrate above the memory cell channel. The cell is programmed by injecting electrons from the cell channel into the nitride, where they are trapped and stored in a limited region. This stored charge then changes the threshold voltage of a portion of the channel of the cell in a manner that is detectable. The cell is erased by injecting hot holes into the nitride. A similar cell can be provided in a split-gate configuration where a doped polysilicon gate extends over a portion of the memory cell channel to form a separate select transistor.

In another approach, NROM cells are used. Two bits, for example, are stored in each NROM cell, where an ONO dielectric layer extends across the channel between source and drain diffusions. The charge for one data bit is localized

in the dielectric layer adjacent to the drain, and the charge for the other data bit localized in the dielectric layer adjacent to the source. Multi-state data storage is obtained by separately reading binary states of the spatially separated charge storage regions within the dielectric. Other types of non-volatile memory are also known.

FIG. 3A illustrates a cross-sectional view of example floating gate memory cells **300**, **310**, **320** in NAND strings. In this Figure, a bit line or NAND string direction goes into the page, and a word line direction goes from left to right. As an example, word line **324** extends across NAND strings which include respective channel regions **306**, **316** and **326**. The memory cell **300** includes a control gate **302**, a floating gate **304**, a tunnel oxide layer **305** and the channel region **306**. The memory cell **310** includes a control gate **312**, a floating gate **314**, a tunnel oxide layer **315** and the channel region **316**. The memory cell **320** includes a control gate **322**, a floating gate **321**, a tunnel oxide layer **325** and the channel region **326**. Each memory cell **300**, **310**, **320** is in a different respective NAND string. An inter-poly dielectric (IPD) layer **328** is also illustrated. The control gates **302**, **312**, **322** are portions of the word line. A cross-sectional view along contact line connector **329** is provided in FIG. 3B.

The control gate **302**, **312**, **322** wraps around the floating gate **304**, **314**, **321**, increasing the surface contact area between the control gate **302**, **312**, **322** and floating gate **304**, **314**, **321**. This results in higher IPD capacitance, leading to a higher coupling ratio which makes programming and erase easier. However, as NAND memory devices are scaled down, the spacing between neighboring cells **300**, **310**, **320** becomes smaller so there is almost no space for the control gate **302**, **312**, **322** and the IPD layer **328** between two adjacent floating gates **302**, **312**, **322**.

As an alternative, as shown in FIGS. 4A and 4B, the flat or planar memory cell **400**, **410**, **420** has been developed in which the control gate **402**, **412**, **422** is flat or planar; that is, it does not wrap around the floating gate and its only contact with the charge storage layer **428** is from above it. In this case, there is no advantage in having a tall floating gate. Instead, the floating gate is made much thinner. Further, the floating gate can be used to store charge, or a thin charge trap layer can be used to trap charge. This approach can avoid the issue of ballistic electron transport, where an electron can travel through the floating gate after tunneling through the tunnel oxide during programming.

FIG. 4A depicts a cross-sectional view of example charge-trapping memory cells **400**, **410**, **420** in NAND strings. The view is in a word line direction of memory cells **400**, **410**, **420** comprising a flat control gate and charge-trapping regions as a two-dimensional example of memory cells **400**, **410**, **420** in the memory cell array **126** of FIG. 1. Charge-trapping memory can be used in NOR and NAND flash memory device. This technology uses an insulator such as an SiN film to store electrons, in contrast to a floating-gate MOSFET technology which uses a conductor such as doped polycrystalline silicon to store electrons. As an example, a word line **424** extends across NAND strings which include respective channel regions **406**, **416**, **426**. Portions of the word line provide control gates **402**, **412**, **422**. Below the word line is an IPD layer **428**, charge-trapping layers **404**, **414**, **421**, polysilicon layers **405**, **415**, **425**, and tunneling layers **409**, **407**, **408**. Each charge-trapping layer **404**, **414**, **421** extends continuously in a respective NAND string. The flat configuration of the control gate can be made thinner than a floating gate. Additionally, the memory cells can be placed closer together.

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FIG. 4B illustrates a cross-sectional view of the structure of FIG. 4A along contact line connector 429. The NAND string 430 includes an SGS transistor 431, example memory cells 400, 433, . . . 0.435, and an SGD transistor 436. Passageways in the IPD layer 428 in the SGS and SGD transistors 431, 436 allow the control gate layers 402 and floating gate layers to communicate. The control gate 402 and floating gate layers may be polysilicon and the tunnel oxide layer may be silicon oxide, for instance. The IPD layer 428 can be a stack of nitrides (N) and oxides (O) such as in a N—O—N—O—N configuration.

The NAND string may be formed on a substrate which comprises a p-type substrate region 455, an n-type well 456 and a p-type well 457. N-type source/drain diffusion regions sd1, sd2, sd3, sd4, sd5, sd6 and sd7 are formed in the p-type well. A channel voltage, V_{ch}, may be applied directly to the channel region of the substrate.

FIG. 5 illustrates an example block diagram of the sense block SB1 of FIG. 1. In one approach, a sense block comprises multiple sense circuits. Each sense circuit is associated with data latches. For example, the example sense circuits 550a, 551a, 552a, and 553a are associated with the data latches 550b, 551b, 552b, and 553b, respectively. In one approach, different subsets of bit lines can be sensed using different respective sense blocks. This allows the processing load which is associated with the sense circuits to be divided up and handled by a respective processor in each sense block. For example, a sense circuit controller 560 in SB1 can communicate with the set of sense circuits and latches. The sense circuit controller 560 may include a pre-charge circuit 561 which provides a voltage to each sense circuit for setting a pre-charge voltage. In one possible approach, the voltage is provided to each sense circuit independently, e.g., via the data bus and a local bus. In another possible approach, a common voltage is provided to each sense circuit concurrently. The sense circuit controller 560 may also include a pre-charge circuit 561, a memory 562 and a processor 563. The memory 562 may store code which is executable by the processor to perform the functions described herein. These functions can include reading the latches 550b, 551b, 552b, 553b which are associated with the sense circuits 550a, 551a, 552a, 553a, setting bit values in the latches and providing voltages for setting pre-charge levels in sense nodes of the sense circuits 550a, 551a, 552a, 553a. Further example details of the sense circuit controller 560 and the sense circuits 550a, 551a, 552a, 553a are provided below.

In some embodiments, a memory cell may include a flag register that includes a set of latches storing flag bits. In some embodiments, a quantity of flag registers may correspond to a quantity of data states. In some embodiments, one or more flag registers may be used to control a type of verification technique used when verifying memory cells. In some embodiments, a flag bit's output may modify associated logic of the device, e.g., address decoding circuitry, such that a specified block of cells is selected. A bulk operation (e.g., an erase operation, etc.) may be carried out using the flags set in the flag register, or a combination of the flag register with the address register, as in implied addressing, or alternatively by straight addressing with the address register alone.

FIG. 6A is a perspective view of a set of blocks 600 in an example three-dimensional configuration of the memory array 126 of FIG. 1. On the substrate are example blocks BLK0, BLK1, BLK2, BLK3 of memory cells (storage elements) and a peripheral area 604 with circuitry for use by the blocks BLK0, BLK1, BLK2, BLK3. For example, the circuitry can include voltage drivers 605 which can be

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connected to control gate layers of the blocks BLK0, BLK1, BLK2, BLK3. In one approach, control gate layers at a common height in the blocks BLK0, BLK1, BLK2, BLK3 are commonly driven. The substrate 601 can also carry circuitry under the blocks BLK0, BLK1, BLK2, BLK3, along with one or more lower metal layers which are patterned in conductive paths to carry signals of the circuitry. The blocks BLK0, BLK1, BLK2, BLK3 are formed in an intermediate region 602 of the memory device. In an upper region 603 of the memory device, one or more upper metal layers are patterned in conductive paths to carry signals of the circuitry. Each block BLK0, BLK1, BLK2, BLK3 comprises a stacked area of memory cells, where alternating levels of the stack represent word lines. In one possible approach, each block BLK0, BLK1, BLK2, BLK3 has opposing tiered sides from which vertical contacts extend upward to an upper metal layer to form connections to conductive paths. While four blocks BLK0, BLK1, BLK2, BLK3 are illustrated as an example, two or more blocks can be used, extending in the x- and/or y-directions.

In one possible approach, the length of the plane, in the x-direction, represents a direction in which signal paths to word lines extend in the one or more upper metal layers (a word line or SGD line direction), and the width of the plane, in the y-direction, represents a direction in which signal paths to bit lines extend in the one or more upper metal layers (a bit line direction). The z-direction represents a height of the memory device.

FIG. 6B illustrates an example cross-sectional view of a portion of one of the blocks BLK0, BLK1, BLK2, BLK3 of FIG. 6A. The block comprises a stack 610 of alternating conductive and dielectric layers. In this example, the conductive layers comprise two SGD layers, two SGS layers and four dummy word line layers DWLD0, DWLD1, DWLS0 and DWLS1, in addition to data word line layers (word lines) WL0-WL111. The dielectric layers are labelled as DL0-DL116. Further, regions of the stack 610 which comprise NAND strings NS1 and NS2 are illustrated. Each NAND string encompasses a memory hole 618, 619 which is filled with materials which form memory cells adjacent to the word lines. A region 622 of the stack 610 is shown in greater detail in FIG. 6D and is discussed in further detail below.

The 610 stack includes a substrate 611, an insulating film 612 on the substrate 611, and a portion of a source line SL. NS1 has a source-end 613 at a bottom 614 of the stack and a drain-end 615 at a top 616 of the stack 610. Contact line connectors (e.g., slits, such as metal-filled slits) 617, 620 may be provided periodically across the stack 610 as interconnects which extend through the stack 610, such as to connect the source line to a particular contact line above the stack 610. The contact line connectors 617, 620 may be used during the formation of the word lines and subsequently filled with metal. A portion of a bit line BL0 is also illustrated. A conductive via 621 connects the drain-end 615 to BL0.

FIG. 6C illustrates a plot of memory hole diameter in the stack of FIG. 6B. The vertical axis is aligned with the stack of FIG. 6B and illustrates a width (wMH), e.g., diameter, of the memory holes 618 and 619. The word line layers WL0-WL111 of FIG. 6A are repeated as an example and are at respective heights z0-z111 in the stack. In such a memory device, the memory holes which are etched through the stack have a very high aspect ratio. For example, a depth-to-diameter ratio of about 25-30 is common. The memory holes may have a circular cross-section. Due to the etching process, the memory hole width can vary along the length of the

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hole. Typically, the diameter becomes progressively smaller from the top to the bottom of the memory hole. That is, the memory holes are tapered, narrowing at the bottom of the stack. In some cases, a slight narrowing occurs at the top of the hole near the select gate so that the diameter becomes slightly wider before becoming progressively smaller from the top to the bottom of the memory hole.

Due to the non-uniformity in the width of the memory hole, the programming speed, including the program slope and erase speed of the memory cells can vary based on their position along the memory hole, e.g., based on their height in the stack. With a smaller diameter memory hole, the electric field across the tunnel oxide is relatively stronger, so that the programming and erase speed is relatively higher. One approach is to define groups of adjacent word lines for which the memory hole diameter is similar, e.g., within a defined range of diameter, and to apply an optimized verify scheme for each word line in a group. Different groups can have different optimized verify schemes.

FIG. 6D illustrates a close-up view of the region 622 of the stack 610 of FIG. 6B. Memory cells are formed at the different levels of the stack at the intersection of a word line layer and a memory hole. In this example, SGD transistors 680, 681 are provided above dummy memory cells 682, 683 and a data memory cell MC. A number of layers can be deposited along the sidewall (SW) of the memory hole 630 and/or within each word line layer, e.g., using atomic layer deposition. For example, each column (e.g., the pillar which is formed by the materials within a memory hole 630) can include a charge-trapping layer or film 663 such as SiN or other nitride, a tunneling layer 664, a polysilicon body or channel 665, and a dielectric core 666. A word line layer can include a blocking oxide/block high-k material 660, a metal barrier 661, and a conductive metal 662 such as Tungsten as a control gate. For example, control gates 690, 691, 692, 693, and 694 are provided. In this example, all of the layers except the metal are provided in the memory hole 630. In other approaches, some of the layers can be in the control gate layer. Additional pillars are similarly formed in the different memory holes. A pillar can form a columnar active area (AA) of a NAND string.

When a memory cell is programmed, electrons are stored in a portion of the charge-trapping layer which is associated with the memory cell. These electrons are drawn into the charge-trapping layer from the channel, and through the tunneling layer. The V_{th} of a memory cell is increased in proportion to the amount of stored charge. During an erase operation, the electrons return to the channel.

Each of the memory holes 630 can be filled with a plurality of annular layers comprising a blocking oxide layer, a charge trapping layer 663, a tunneling layer 664 and a channel layer. A core region of each of the memory holes 630 is filled with a body material, and the plurality of annular layers are between the core region and the word line in each of the memory holes 630.

The NAND string can be considered to have a floating body channel because the length of the channel is not formed on a substrate. Further, the NAND string is provided by a plurality of word line layers above one another in a stack, and separated from one another by dielectric layers.

FIG. 7A illustrates a top view of an example word line layer WL0 of the stack 610 of FIG. 6B. As mentioned, a three-dimensional memory device can comprise a stack of alternating conductive and dielectric layers. The conductive layers provide the control gates of the SG transistors and memory cells. The layers used for the SG transistors are SG layers and the layers used for the memory cells are word line

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layers. Further, memory holes are formed in the stack and filled with a charge-trapping material and a channel material. As a result, a vertical NAND string is formed. Source lines are connected to the NAND strings below the stack and bit lines are connected to the NAND strings above the stack.

A block BLK in a three-dimensional memory device can be divided into sub-blocks, where each sub-block comprises a NAND string group which has a common SGD control line. For example, see the SGD lines/control gates SGD0, SGD1, SGD2 and SGD3 in the sub-blocks SBa, SBb, SBc and SBd, respectively. Further, a word line layer in a block can be divided into regions. Each region is in a respective sub-block and can extend between contact line connectors (e.g., slits) which are formed periodically in the stack to process the word line layers during the fabrication process of the memory device. This processing can include replacing a sacrificial material of the word line layers with metal. Generally, the distance between contact line connectors should be relatively small to account for a limit in the distance that an etchant can travel laterally to remove the sacrificial material, and that the metal can travel to fill a void which is created by the removal of the sacrificial material. For example, the distance between contact line connectors may allow for a few rows of memory holes between adjacent contact line connectors. The layout of the memory holes and contact line connectors should also account for a limit in the number of bit lines which can extend across the region while each bit line is connected to a different memory cell. After processing the word line layers, the contact line connectors can optionally be filled with metal to provide an interconnect through the stack.

In this example, there are four rows of memory holes between adjacent contact line connectors. A row here is a group of memory holes which are aligned in the x-direction. Moreover, the rows of memory holes are in a staggered pattern to increase the density of the memory holes. The word line layer or word line is divided into regions WL0a, WL0b, WL0c and WL0d which are each connected by a contact line 713. The last region of a word line layer in a block can be connected to a first region of a word line layer in a next block, in one approach. The contact line 713, in turn, is connected to a voltage driver for the word line layer. The region WL0a has example memory holes 710, 711 along a contact line 712. The region WL0b has example memory holes 714, 715. The region WL0c has example memory holes 716, 717. The region WL0d has example memory holes 718, 719. The memory holes are also shown in FIG. 7B. Each memory hole can be part of a respective NAND string. For example, the memory holes 710, 714, 716 and 718 can be part of NAND strings NS0_SBa, NS1_SBB, NS2_SBc, NS3_SBd, and NS4_SBe, respectively.

Each circle represents the cross-section of a memory hole at a word line layer or SG layer. Example circles shown with dashed lines represent memory cells which are provided by the materials in the memory hole and by the adjacent word line layer. For example, memory cells 720, 721 are in WL0a, memory cells 724, 725 are in WL0b, memory cells 726, 727 are in WL0c, and memory cells 728, 729 are in WL0d. These memory cells are at a common height in the stack.

Contact line connectors (e.g., slits, such as metal-filled slits) 701, 702, 703, 704 may be located between and adjacent to the edges of the regions WL0a-WL0d. The contact line connectors 701, 702, 703, 704 provide a conductive path from the bottom of the stack to the top of the stack. For example, a source line at the bottom of the stack may be connected to a conductive line above the stack,

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where the conductive line is connected to a voltage driver in a peripheral region of the memory device.

FIG. 7B illustrates a top view of an example top dielectric layer DL116 of the stack of FIG. 6B. The dielectric layer is divided into regions DL116a, DL116b, DL116c and DL116d. Each region can be connected to a respective voltage driver. This allows a set of memory cells in one region of a word line layer being programmed concurrently, with each memory cell being in a respective NAND string which is connected to a respective bit line. A voltage can be set on each bit line to allow or inhibit programming during each program voltage.

The region DL116a has the example memory holes 710, 711 along a contact line 712, which is coincident with a bit line BL0. A number of bit lines extend above the memory holes and are connected to the memory holes as indicated by the "X" symbols. BL0 is connected to a set of memory holes which includes the memory holes 711, 715, 717, 719. Another example bit line BL1 is connected to a set of memory holes which includes the memory holes 710, 714, 716, 718. The contact line connectors (e.g., slits, such as metal-filled slits) 701, 702, 703, 704 from FIG. 7A are also illustrated, as they extend vertically through the stack. The bit lines can be numbered in a sequence BL0-BL23 across the DL116 layer in the x-direction.

Different subsets of bit lines are connected to memory cells in different rows. For example, BL0, BL4, BL8, BL12, BL16, BL20 are connected to memory cells in a first row of cells at the right-hand edge of each region. BL2, BL6, BL10, BL14, BL18, BL22 are connected to memory cells in an adjacent row of cells, adjacent to the first row at the right-hand edge. BL3, BL7, BL11, BL15, BL19, BL23 are connected to memory cells in a first row of cells at the left-hand edge of each region. BL1, BL5, BL9, BL13, BL17, BL21 are connected to memory cells in an adjacent row of memory cells, adjacent to the first row at the left-hand edge.

The memory cells of the memory blocks can be programmed to retain one or more bits of data in multiple data states, each of which is associated with a respective threshold voltage V_t range. For example, FIG. 8 depicts a threshold voltage V_t distribution of a group of memory cells programmed according to a one bit per memory cell (SLC) storage scheme. In the SLC storage scheme, there are two total data states, including the erased state (Er) and a single programmed data state (S1). FIG. 9 illustrates the threshold voltage V_t distribution of a three bits per cell (TLC) storage scheme that includes eight total data states, namely the erased state (Er) and seven programmed data states (S1-S7). Each programmed data state (S1-S7) is associated with a respective verify voltage (V_{v1} - V_{v7}), which is employed during a verify portion of a programming operation. FIG. 10 depicts a threshold voltage V_t distribution of a four bits per cell (QLC) storage scheme that includes sixteen total data states, namely the erased state (Er) and fifteen programmed data states (S1-S15). Other storage schemes are also available, such as two bits per cell (MLC) with four data states or five bits per cell (PLC) with thirty-two data states.

Programming the memory cells of a selected word line to retain multiple bits per memory cell (for example, MLC, TLC, or QLC) typically includes a plurality of program loops. FIG. 11 depicts a waveform 1100 of the voltages applied to a selected word line during an example memory cell programming operation for programming the memory cells of the selected word line to a greater number of bits per memory cell (e.g., TLC or QLC). As depicted, each program loop includes a programming pulse VPGM and one or more

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verify pulses, depending on which data states are being programmed in a particular program loop. A square waveform is depicted for each pulse for simplicity; however, other shapes are possible, such as a multilevel shape or a ramped shape.

Incremental Step Pulse Programming (ISPP) is used in this example pulse train, which means that the VPGM pulse amplitude steps up, or increases, in each successive program loop. More specifically, the pulse train includes VPGM pulses that increase stepwise in amplitude with each successive program loop by a fixed step size (dVPGM). A new pulse train starts with an initial VPGM pulse level VPGMU and ends at a final VPGM pulse level, which does not exceed a maximum allowed level. The example pulse train 1100 includes a series of VPGM pulses 1101-1115 that are applied to a selected word line that includes a set of non-volatile memory cells. One or more verify voltage pulses 1116-1129 are provided after each VPGM pulse as an example, based on the target data states which are being verified in the program loop. The verify voltages correspond with voltages V_{v1} - V_{v7} shown in FIG. 9 or V_{v1} - V_{v15} shown in FIG. 10. Concurrent with the application of the verify voltages, a sensing operation can determine whether a particular memory cell in the selected word line has a threshold voltage V_t above the verify voltage associated with its intended data state by sensing a current through the memory cell. If the current is relatively high during sensing, this indicates that the memory cell is in a conductive state, such that its threshold voltage V_t is less than the verify voltage. If the current is relatively low during sensing, this indicates that the memory cell is in a non-conductive state, such that its threshold voltage V_t is above the verify voltage. If the memory cell passes verify, programming of that memory cell is completed and further programming of that memory cell is inhibited for all remaining program loops by applying an inhibit voltage to a bit line coupled with the memory cell concurrent with the VPGM pulse. Programming proceeds until all memory cells pass verify for their intended data states, in which case, programming passes, or until a predetermined maximum number of program loops is exceeded, in which case, programming fails. In some embodiments, the memory cells of a word line can be divided into a series of string groups, or simply strings, that can be programmed independently of one another, and programming can commence from one string to another across the word line before proceeding to the next word line in the memory block.

Programming of the memory cells of a selected word line starts with all of the memory cells being in the erased data state and can be conducted in either a full sequence programming operation or a multi-pass programming operation. In a full sequence programming operation, the memory cells are programmed directly to their final threshold voltages in a single programming pass, e.g., the programming pass depicted in FIG. 11. One type of multi-pass programming operation is sometimes known as a foggy-fine operation. In a foggy-fine operation, the memory cells are programmed to their final programmed data states in two or more programming passes or stages, e.g., a first (hereinafter referred to as "foggy") pass and a second (hereinafter referred to as "fine") pass. The pulse train of each of these programming passes includes a plurality of program loops with programming pulses and verify pulses and may resemble the programming pass depicted in FIG. 11 but the step voltage dVPGM is greater during the foggy pass than during the fine pass.

In some embodiments, when programming the memory cells of a word line to QLC using a foggy-fine operation,

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during the foggy pass, all data states S1-S15 are verified. In an effort to improve performance during programming (i.e., reduce programming time tProg), according to some programming techniques sometimes known as “reduced verify,” only some of the data states are verified during the foggy pass. Specifically, during the foggy pass, the memory cells of the word line are only verified at certain checkpoint data states and verify is skipped for the non-checkpoint data states to reduce total programming time during the foggy pass. In the example of FIG. 12, there are seven checkpoint data states, which are the even data states (S2, S4, S6, S8, S10, S12, and S14), and verify is skipped for the memory cells being programmed to the odd data states (S1, S3, S5, S7, S9, S11, S13, and S15). In this Figure, the cells with the “Is” identify which program loops which data states are being verified during an example foggy pass. When a memory cell being programmed to an odd data state passes verify for the previous data state, then it receives a predetermined number of “blind” (without verify) programming pulses and then is inhibited from further programming. For example, a memory cell being programmed to data state S3 might receive two blind programming pulses after that memory cell passes verify at data state S2. The blind programming pulse(s) could utilize a quick pass write (QPW) technique where a low voltage is applied to a bit line to slow programming. For example, a memory cell being programmed to the S7 data state might receive two full speed programming pulses and one QPW programming pulse after passing verify at data state S4.

In the example of FIG. 13, there are four checkpoint data states, which are data states S2, S4, S8, and S12, and verify is skipped for the memory cells being programmed to data states S1, S3, S5, S6, S7, S9, S10, S11, S13, S14, and S15. In some embodiments, the number of checkpoint data states and the specific data states that serve as those checkpoint data states could vary from these specific examples.

While reduced verify may improve performance during the foggy programming pass, the data states being programmed to the non-checkpoint data states end up with wider threshold voltage Vt distributions. For example, FIG. 14 depicts a threshold voltage distribution of a page of memory cells following the foggy pass with seven checkpoint data states. As illustrated, the Vt distributions for the odd data states, which are not checkpoint data states, are all wider than the Vt distributions for the even data states, which are checkpoint data states. Also, the number of blind programming pulses to apply to the memory cells being programmed to non-checkpoint data states can be complicated to determine, especially when the programming step size is subject to change. Both of these drawbacks can result in increased failed bit counts (FBC).

One issue with foggy fine operations is that, in some cases, the fine programming pass might not follow immediately or nearly immediately after the foggy pass. Because the threshold voltage distributions are much wider after the foggy pass than after the fine pass such that there may be overlap between the threshold voltage distributions of adjacent data states, it may not be possible to directly read the data from a word line that has only received the foggy pass. Therefore, in some techniques, a safe copy of all the pages of final data for the word line is saved elsewhere in the memory device (for example, in latches), which requires resources in the memory device that could otherwise be used for other purposes. Some of these resources can be freed up by utilizing an encoded foggy-fine technique.

In an encoded foggy-fine technique, rather than storing the entire data elsewhere, during or immediately after the

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foggy pass, one page of parity data (one bit per memory cell in the selected word line) is stored in a reduced number of data states within the memory device, e.g., in NAND or DRAM. In the embodiment of FIG. 15, each parity bit is either in the erased data state Er or in an “A” data state, i.e., the parity data is in an SLC storage scheme. As illustrated in FIG. 15, prior to the fine pass, the foggy data 1500 and the page of parity data 1502 can be read to reconstruct the final data, which can then be programmed into the word line to produce the final threshold voltage distribution 1504 during the fine pass. For example, when a memory cell is read as being in the range of either the S2 or S3 data state, then the parity bit associated with that memory cell is referenced to determine which of these two data states S2, S3 is the intended final data state for that memory cell. In the parity data, the erased data state Er can be associated with either the odd final data states and the A data state can be associated with the other or vice versa.

While these encoded foggy-fine techniques allow the memory device to reconstruct the final data where the threshold voltage Vt curves of neighboring data states overlap with one another, if there is more than a small amount of overlap between data states that are two data states away from one another (e.g., S1 and S3 or S2 and S4), this can result in an elevated failed bit count (FBC). Also, such encoded foggy-fine techniques where the parity data is stored in SLC format are not compatible with reduced verify techniques that include less than seven checkpoint data states.

According to one aspect of the present disclosure, a programming technique is provided to improve the reliability of reduced verify and encoded foggy-fine programming techniques with little or no reduction in performance. Specifically, as illustrated in FIG. 16B, the parity data contains three possible data states, namely, the erased data state Er, the A data state, and the B data state. In other words, each memory cell in the word line receives parity data that can either be in the erased data state Er, the A data state, or the B data state. As illustrated between FIGS. 16A and 16B, for each group of three QLC data states (e.g., the group of Er, S1, and S2 or the group of S3, S4, and S5), each of these parity data states is associated with one bit in the group. In the illustrated example, the erased data state Er of the parity data is associated with the Er, S3, S6, S9, S12, and S15 data states; the A data state of the parity data is associated with the S1, S4, S7, S10, and S13 data states; and the B data state of the parity data is associated with the S2, S5, S8, S11, and S14 data states.

Prior to the fine pass, the control circuitry of the memory device reads the foggy data and the parity data and reconstructs the final data. For each memory cell, the control circuitry determines which group it is in from the foggy data and which data state within that group from the parity data. Without erasing the foggy data, the control circuitry then performs the fine pass to tighten the threshold voltage Vt distributions for all of the programmed data states S1-S15 to achieve the desired final threshold voltage Vt distribution with very accurately programmed data and little or no overlap between adjacent data states, e.g., see FIG. 10.

By including three possible data states in the parity data, the memory device is able to distinguish between memory cells where their threshold distributions might overlap with data states that are two away from one another, e.g. data state S(N) with data state S(N±2).

If the parity data is stored in NAND, then it can (optionally) be programmed in a 2P0V programming operation that includes two programming pulses and zero verify opera-

tions. When reading the parity data, only two sensing operations need to be performed: one for the A data state and one for the B data state.

If the parity data is stored in DRAM, then the parity data needs to be stored as two bits per memory cell, similar to MLC, but only three of the possible four data states are utilized, e.g., [00, 01, 11] or [00, 10, 11]. Thus, one of the DRAM memory cells will be programmed less frequently than the other, improving the overall endurance of the DRAM.

The three data state parity data can be utilized in combination with a range of different types of foggy-fine programming operations, including fifteen checkpoint foggy-fine, seven checkpoint reduced verify, and five checkpoint reduced verify, as discussed below.

Turning now to FIGS. 16A, 16B, and 17, in this example embodiment, during the foggy pass, the verify operation is performed at five checkpoint data states, namely S1, S4, S7, S10 and S13, and verify is skipped for the remaining programmed data states S2, S3, S5, S6, S8, S9, S11, S12, S14 and S15, while Er state remains untouched. This divides the programmed data states into five groups of three data states. A first group of three data states includes data states S1, S2 and S3. A second group of three data states includes data states S4, S5 and S6. A third group of three data states includes data states S7, S8 and S9. A fourth group of three data states includes data states S10, S11 and S12. A fifth group of three data states includes data states S13, S14 and S15. In each of the five groups of three data states, one data state is associated with each of the three data states of the parity data. For example, in the group S1, S2 and S3, the S3 data state of the foggy distribution can be associated with the 'Er' data state of the parity data, the S1 data state of the foggy distribution can be associated with the 'A' data state of the parity data, and the S2 data state of the foggy distribution can be associated with the 'B' data state of the parity data.

Prior to the fine pass, the control circuitry can read the foggy data 1600 and the parity data 1602 to reconstruct the final data and then perform the fine pass to achieve the final threshold voltage V_t distribution 1700.

Five checkpoint foggy-fine programming in combination with three data state parity data results in an improved encoded foggy-fine programming technique that reduces system performance overhead without compromising on data reliability as compared to other known foggy-fine approaches. Whether used in combination with five checkpoint foggy-fine programming or with another type of foggy-fine programming, the three data state parity data also results in significantly reduced fail bit counts as compared to other encoded foggy-fine approaches.

Turning now to FIG. 18, a flow chart is provided depicting the steps of programming a selected word line according to an example embodiment of the present disclosure. At step 1800, a request to write user data to a selected word line WLn is received. At step 1802, control circuitry in the memory device performs a programming pass to write the foggy data to the selected word line WLn at high speed. In one presently preferred embodiment, the programming pass that programs the foggy data to the selected word line WLn is a reduced verify operation with five checkpoint data states. At step 1804, the control circuitry programs the parity data into the memory device somewhere other than in the selected word line WLn , for example, elsewhere in NAND or in DRAM. The parity data includes three possible data states. At this point, there is no need to retain a safe copy of the user data and any such copy can be erased.

There may be an indeterminate and long gap between steps 1804 and 1806. At steps 1806 and 1808, the control circuitry reads the foggy data from the selected word line WLn and reads the parity data. At step 1810, the control circuitry reconstructs the user data from the foggy data and the parity data. Due to the wide threshold voltage V_t distribution curves in the foggy data some curves of neighboring data states may overlap with one another. The parity data is able to determine which particular one of up to two overlapping data states the memory cell is intended to be in. At step 1812, the control circuitry performs the fine pass on the selected word line WLn to more accurately program the memory cells of the selected word line from the foggy data to the user data. Programming can then move on to a next word line.

In some embodiments, the programming techniques discussed above in reference to QLC programming could also be applied to TLC programming.

Various terms are used herein to refer to particular system components. Different companies may refer to a same or similar component by different names and this description does not intend to distinguish between components that differ in name but not in function. To the extent that various functional units described in the following disclosure are referred to as "modules," such a characterization is intended to not unduly restrict the range of potential implementation mechanisms. For example, a "module" could be implemented as a hardware circuit that includes customized very-large-scale integration (VLSI) circuits or gate arrays, or off-the-shelf semiconductors that include logic chips, transistors, or other discrete components. In a further example, a module may also be implemented in a programmable hardware device such as a field programmable gate array (FPGA), programmable array logic, a programmable logic device, or the like. Furthermore, a module may also, at least in part, be implemented by software executed by various types of processors. For example, a module may comprise a segment of executable code constituting one or more physical or logical blocks of computer instructions that translate into an object, process, or function. Also, it is not required that the executable portions of such a module be physically located together, but rather, may comprise disparate instructions that are stored in different locations and which, when executed together, comprise the identified module and achieve the stated purpose of that module. The executable code may comprise just a single instruction or a set of multiple instructions, as well as be distributed over different code segments, or among different programs, or across several memory devices, etc. In a software, or partial software, module implementation, the software portions may be stored on one or more computer-readable and/or executable storage media that include, but are not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor-based system, apparatus, or device, or any suitable combination thereof. In general, for purposes of the present disclosure, a computer-readable and/or executable storage medium may be comprised of any tangible and/or non-transitory medium that is capable of containing and/or storing a program for use by or in connection with an instruction execution system, apparatus, processor, or device.

Similarly, for the purposes of the present disclosure, the term "component" may be comprised of any tangible, physical, and non-transitory device. For example, a component may be in the form of a hardware logic circuit that is comprised of customized VLSI circuits, gate arrays, or other integrated circuits, or is comprised of off-the-shelf semicon-

ductors that include logic chips, transistors, or other discrete components, or any other suitable mechanical and/or electronic devices. In addition, a component could also be implemented in programmable hardware devices such as field programmable gate arrays (FPGA), programmable array logic, programmable logic devices, etc. Furthermore, a component may be comprised of one or more silicon-based integrated circuit devices, such as chips, die, die planes, and packages, or other discrete electrical devices, in an electrical communication configuration with one or more other components via electrical conductors of, for example, a printed circuit board (PCB) or the like. Accordingly, a module, as defined above, may in certain embodiments, be embodied by or implemented as a component and, in some instances, the terms module and component may be used interchangeably.

Where the term "circuit" is used herein, it includes one or more electrical and/or electronic components that constitute one or more conductive pathways that allow for electrical current to flow. A circuit may be in the form of a closed-loop configuration or an open-loop configuration. In a closed-loop configuration, the circuit components may provide a return pathway for the electrical current. By contrast, in an open-looped configuration, the circuit components therein may still be regarded as forming a circuit despite not including a return pathway for the electrical current. For example, an integrated circuit is referred to as a circuit irrespective of whether the integrated circuit is coupled to ground (as a return pathway for the electrical current) or not. In certain exemplary embodiments, a circuit may comprise a set of integrated circuits, a sole integrated circuit, or a portion of an integrated circuit. For example, a circuit may include customized VLSI circuits, gate arrays, logic circuits, and/or other forms of integrated circuits, as well as may include off-the-shelf semiconductors such as logic chips, transistors, or other discrete devices. In a further example, a circuit may comprise one or more silicon-based integrated circuit devices, such as chips, die, die planes, and packages, or other discrete electrical devices, in an electrical communication configuration with one or more other components via electrical conductors of, for example, a printed circuit board (PCB). A circuit could also be implemented as a synthesized circuit with respect to a programmable hardware device such as a field programmable gate array (FPGA), programmable array logic, and/or programmable logic devices, etc. In other exemplary embodiments, a circuit may comprise a network of non-integrated electrical and/or electronic components (with or without integrated circuit devices). Accordingly, a module, as defined above, may in certain embodiments, be embodied by or implemented as a circuit.

It will be appreciated that example embodiments that are disclosed herein may be comprised of one or more microprocessors and particular stored computer program instructions that control the one or more microprocessors to implement, in conjunction with certain non-processor circuits and other elements, some, most, or all of the functions disclosed herein. Alternatively, some or all functions could be implemented by a state machine that has no stored program instructions, or in one or more application-specific integrated circuits (ASICs) or field-programmable gate arrays (FPGAs), in which each function or some combinations of certain of the functions are implemented as custom logic. A combination of these approaches may also be used. Further, references below to a "controller" shall be defined as comprising individual circuit components, an application-specific integrated circuit (ASIC), a microcontroller with controlling software, a digital signal processor (DSP), a field

programmable gate array (FPGA), and/or a processor with controlling software, or combinations thereof.

Additionally, the terms "couple," "coupled," or "couples," where may be used herein, are intended to mean either a direct or an indirect connection. Thus, if a first device couples, or is coupled to, a second device, that connection may be by way of a direct connection or through an indirect connection via other devices (or components) and connections.

Regarding, the use herein of terms such as "an embodiment," "one embodiment," an "exemplary embodiment," a "particular embodiment," or other similar terminology, these terms are intended to indicate that a specific feature, structure, function, operation, or characteristic described in connection with the embodiment is found in at least one embodiment of the present disclosure. Therefore, the appearances of phrases such as "in one embodiment," "in an embodiment," "in an exemplary embodiment," etc., may, but do not necessarily, all refer to the same embodiment, but rather, mean "one or more but not all embodiments" unless expressly specified otherwise. Further, the terms "comprising," "having," "including," and variations thereof, are used in an open-ended manner and, therefore, should be interpreted to mean "including, but not limited to . . ." unless expressly specified otherwise. Also, an element that is preceded by "comprises . . . a" does not, without more constraints, preclude the existence of additional identical elements in the subject process, method, system, article, or apparatus that includes the element.

The terms "a," "an," and "the" also refer to "one or more" unless expressly specified otherwise. In addition, the phrase "at least one of A and B" as may be used herein and/or in the following claims, whereby A and B are variables indicating a particular object or attribute, indicates a choice of A or B, or both A and B, similar to the phrase "and/or." Where more than two variables are present in such a phrase, this phrase is hereby defined as including only one of the variables, any one of the variables, any combination (or sub-combination) of any of the variables, and all of the variables.

Further, where used herein, the term "about" or "approximately" applies to all numeric values, whether or not explicitly indicated. These terms generally refer to a range of numeric values that one of skill in the art would consider equivalent to the recited values (e.g., having the same function or result). In certain instances, these terms may include numeric values that are rounded to the nearest significant figure.

In addition, any enumerated listing of items that is set forth herein does not imply that any or all of the items listed are mutually exclusive and/or mutually inclusive of one another, unless expressly specified otherwise. Further, the term "set," as used herein, shall be interpreted to mean "one or more," and in the case of "sets," shall be interpreted to mean multiples of (or a plurality of) "one or more," "ones or more," and/or "ones or mores" according to set theory, unless expressly specified otherwise.

The foregoing detailed description has been presented for purposes of illustration and description. It is not intended to be exhaustive or be limited to the precise form disclosed. Many modifications and variations are possible in light of the above description. The described embodiments were chosen to best explain the principles of the technology and its practical application to thereby enable others skilled in the art to best utilize the technology in various embodiments and with various modifications as are suited to the particular use contemplated. The scope of the technology is defined by the claims appended hereto.

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What is claimed is:

1. A method of programming a memory device, comprising the steps of:

preparing a memory block that includes an array of memory cells arranged in a plurality of word lines;

programming final data into a selected word line of the plurality of word lines in a multi-pass programming operation that includes the steps of:

in a first pass, programming foggy data to the memory cells of the selected word line and programming parity data in the memory device, the parity data including three possible data states,

prior to a second pass, reading the foggy data and the parity data and reconstructing the final data from the foggy data and the parity data, and

in a second pass, programming the memory cells of the selected word line from the foggy data to the final data.

2. The method as set forth in claim 1, wherein the final data includes four bits per memory cell of the selected word line with each memory cell either remaining in an erased data state or being programmed to one of fifteen programmed data states.

3. The method as set forth in claim 2, wherein the first pass includes a plurality of program loops, each program loop including a programming pulse and a verify operation, and wherein the verify operation is performed at a plurality of checkpoint data states, the plurality of checkpoint data states being less than the fifteen programmed data states.

4. The method as set forth in claim 3, wherein the plurality of checkpoint data states includes exactly five checkpoint data states.

5. The method as set forth in claim 3, wherein the plurality of checkpoint data states includes more than five checkpoint data states.

6. The method as set forth in claim 1, wherein the memory cells being programmed to non-checkpoint data states are verified at a previous one of the plurality of checkpoint data states and then are programmed in at least one program loop that includes a programming pulse and no verify operation.

7. The method as set forth in claim 1, wherein the page of parity data is stored in NAND or DRAM.

8. A memory device, comprising:

a memory block that includes an array of memory cells arranged in a plurality of word lines;

control circuitry that is configured to program final data into a selected word line of the plurality of word lines in a multi-pass programming operation, while programming the final data to the selected word line, the control circuitry being configured to:

in a first pass, program the memory cells of the selected word line to foggy data and program parity data in the memory device, the parity data including three possible data states,

prior to a second pass, read the foggy data and the parity data and reconstruct the final data from the foggy data and the parity data, and

in a second pass, program the memory cells of the selected word line from the foggy data to the final data.

9. The memory device as set forth in claim 8, wherein the final data includes four bits per memory cell of the selected word line with each memory cell either remaining in an erased data state or being programmed to one of fifteen programmed data states.

10. The memory device as set forth in claim 9, wherein the first pass includes a plurality of program loops, in each

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program loop the control circuitry is configured to apply a programming pulse and conduct a verify operation to the selected word line, and wherein the verify operation is performed by the control circuitry at a plurality of checkpoint data states, the plurality of checkpoint data states being less than the fifteen programmed data states.

11. The memory device as set forth in claim 10, wherein the plurality of checkpoint data states includes exactly five checkpoint data states.

12. The memory device as set forth in claim 10, wherein the plurality of checkpoint data states includes more than five checkpoint data states.

13. The memory device as set forth in claim 8, wherein the control circuitry is configured to verify the memory cells being programmed to non-checkpoint data states at a previous one of the plurality of checkpoint data states and then program those memory cells being programmed to non-checkpoint data states in at least one program loop that includes a programming pulse and no verify operation.

14. The memory device as set forth in claim 8, wherein the control circuitry is configured to store the page of parity data in NAND or DRAM.

15. An apparatus, comprising:

a memory block that includes an array of memory cells arranged in a plurality of word lines;

a programming means for programming final data into a selected word line of the plurality of word lines in a multi-pass programming operation, while programming the final data to the selected word line, the programming means is configured to:

in a first pass, program the memory cells of the selected word line to foggy data and program corresponding parity data in the memory device, the parity data including three possible data states,

prior to a second pass, read the foggy data and the parity data and reconstruct the final data from the foggy data and the parity data, and

in a second pass, program the memory cells of the selected word line from the foggy data to the final data.

16. The apparatus as set forth in claim 15, wherein the final data includes four bits per memory cell of the selected word line with each memory cell either remaining in an erased data state or being programmed to one of fifteen programmed data states.

17. The apparatus as set forth in claim 16, wherein the first pass includes a plurality of program loops, in each program loop the programming means is configured to apply a programming pulse and a verify operation to the selected word line, and wherein the verify operation is performed by the programming means at a plurality of checkpoint data states, the plurality of checkpoint data states being less than the fifteen programmed data states.

18. The apparatus as set forth in claim 17, wherein the plurality of checkpoint data states includes exactly five checkpoint data states.

19. The apparatus as set forth in claim 17, wherein the plurality of checkpoint data states includes more than five checkpoint data states.

20. The apparatus as set forth in claim 15, wherein the programming means is configured to verify the memory cells being programmed to non-checkpoint data states at a previous one of the plurality of checkpoint data states and then program those memory cells being programmed to

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non-checkpoint data states in at least one program loop that includes a programming pulse and no verify operation.

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